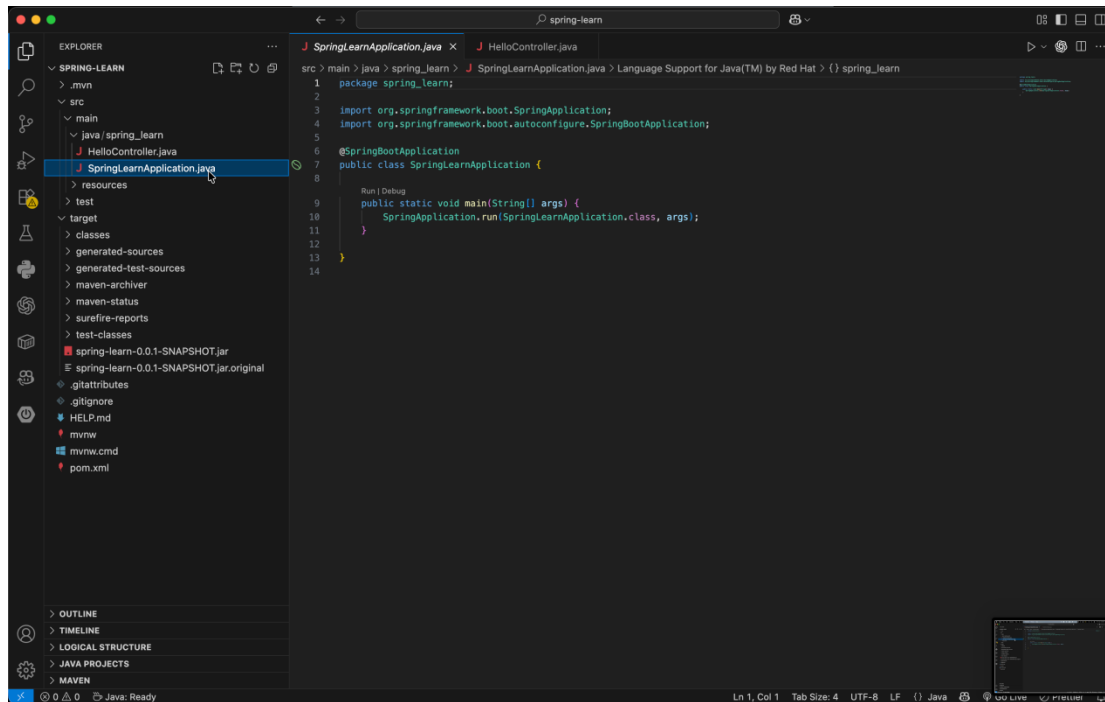
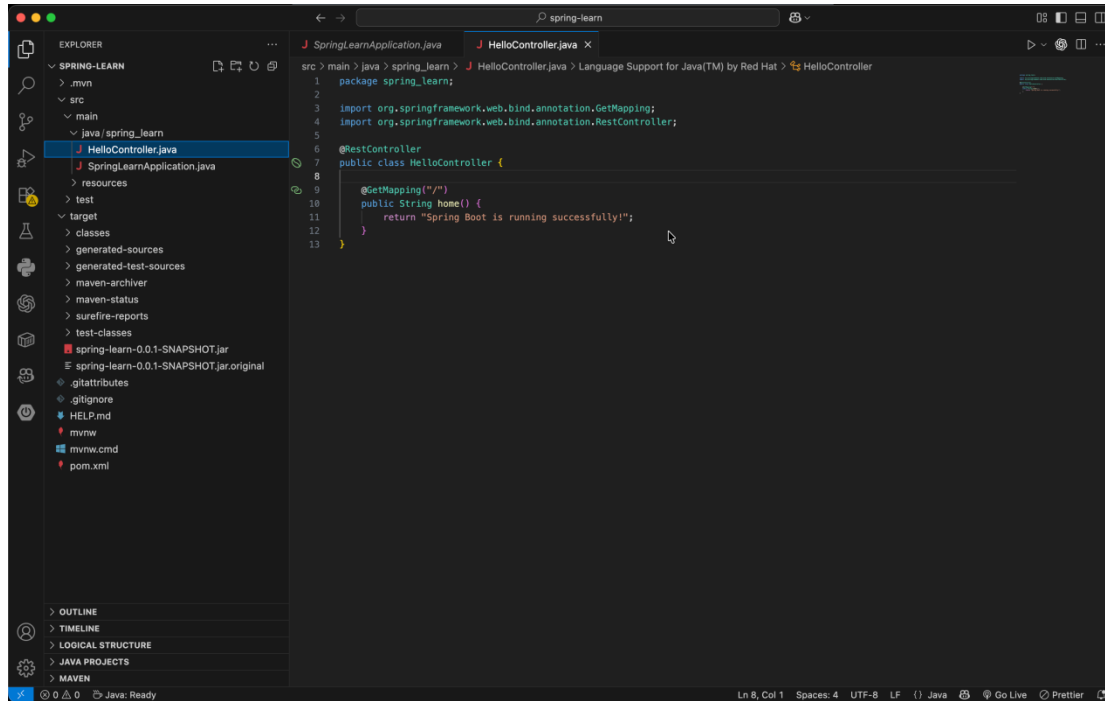
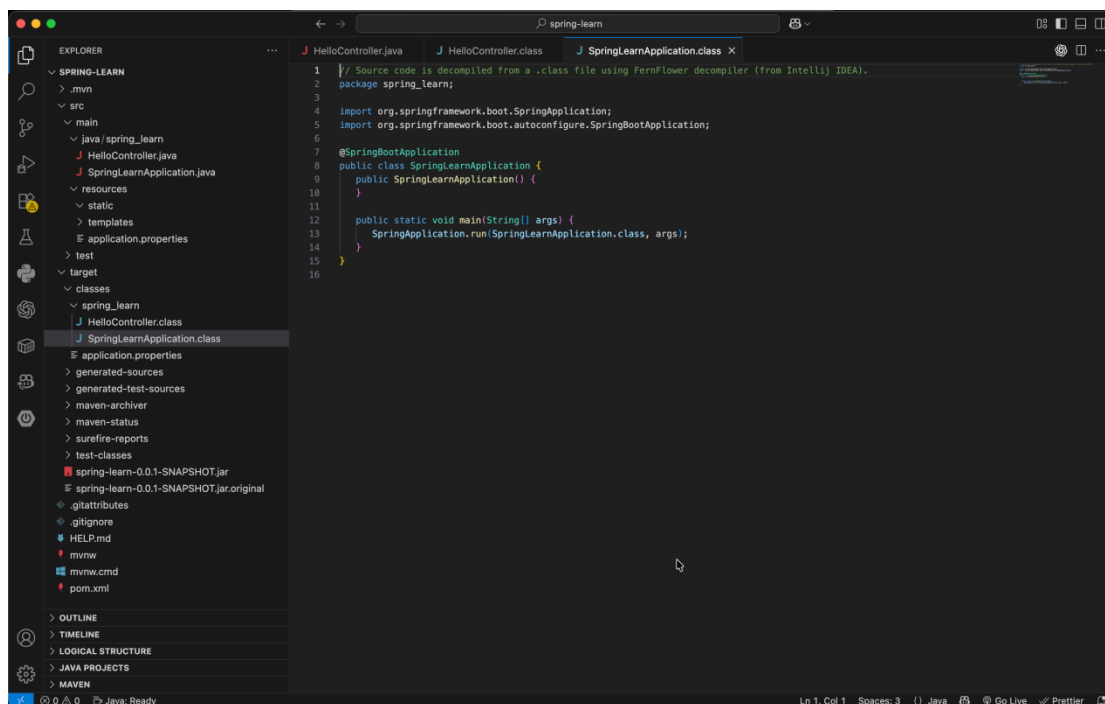
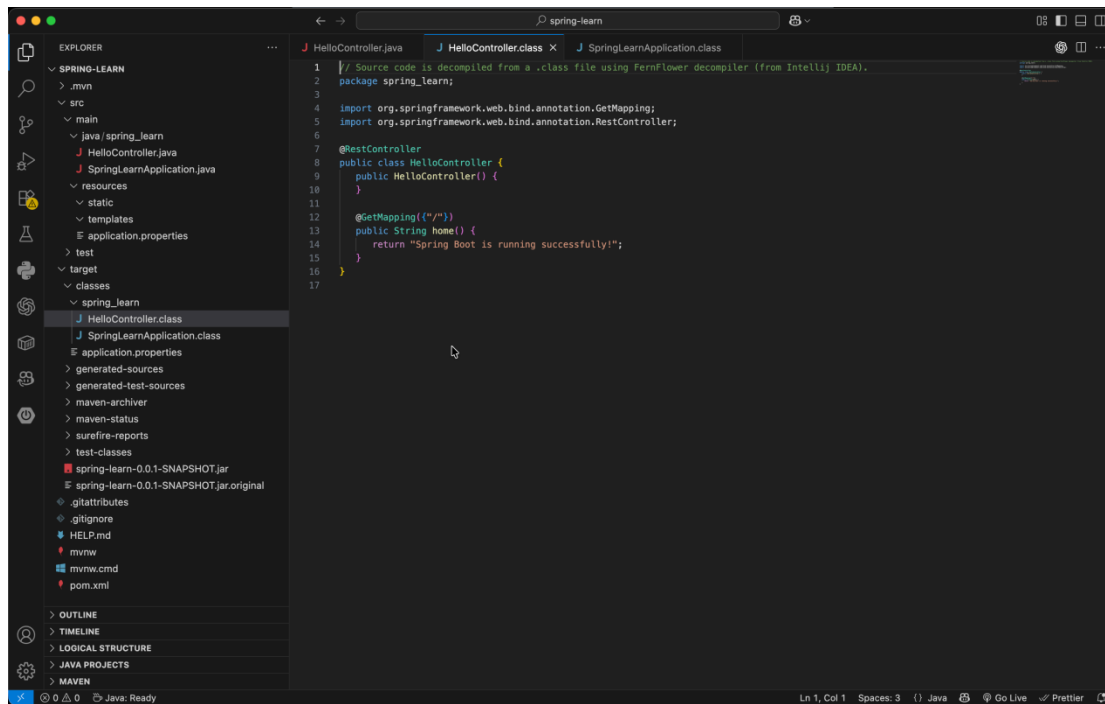


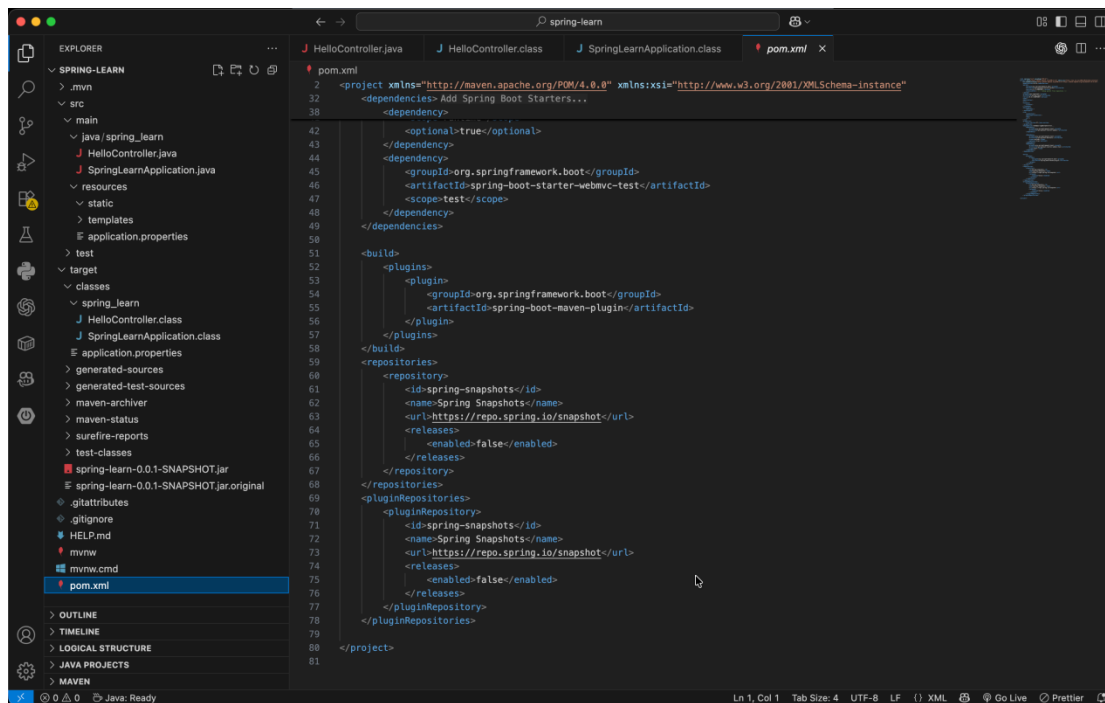
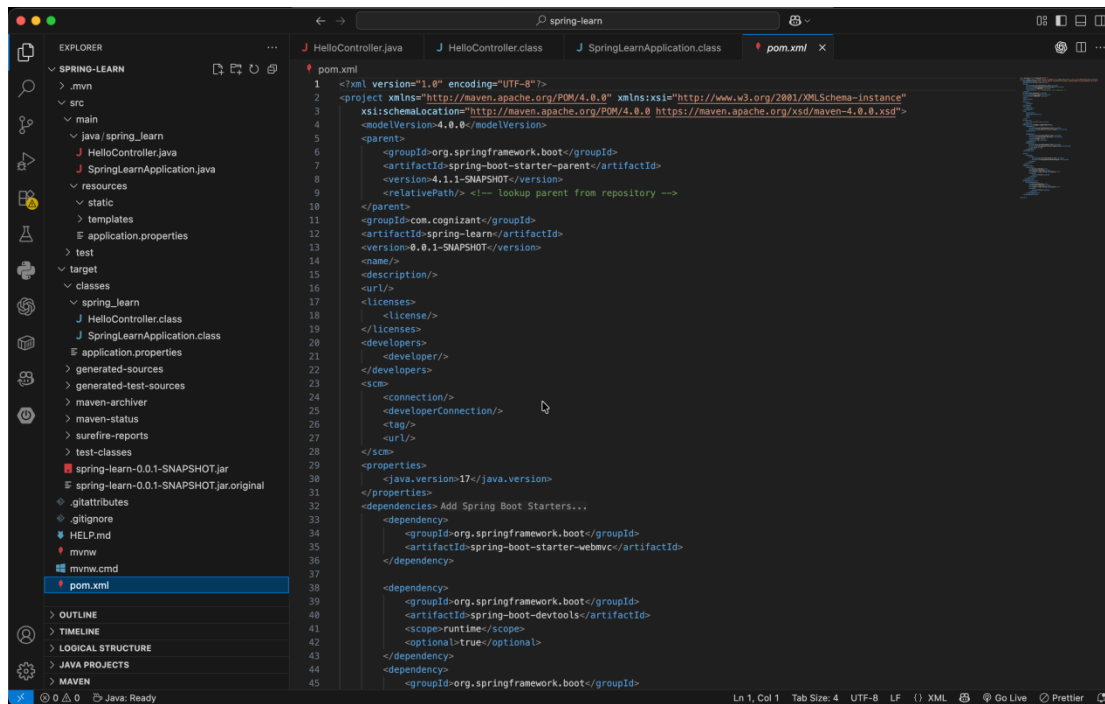
DIGITAL NURTURE 5.0 DEEPSKILLING

WEEK 3 – HANDS-ON

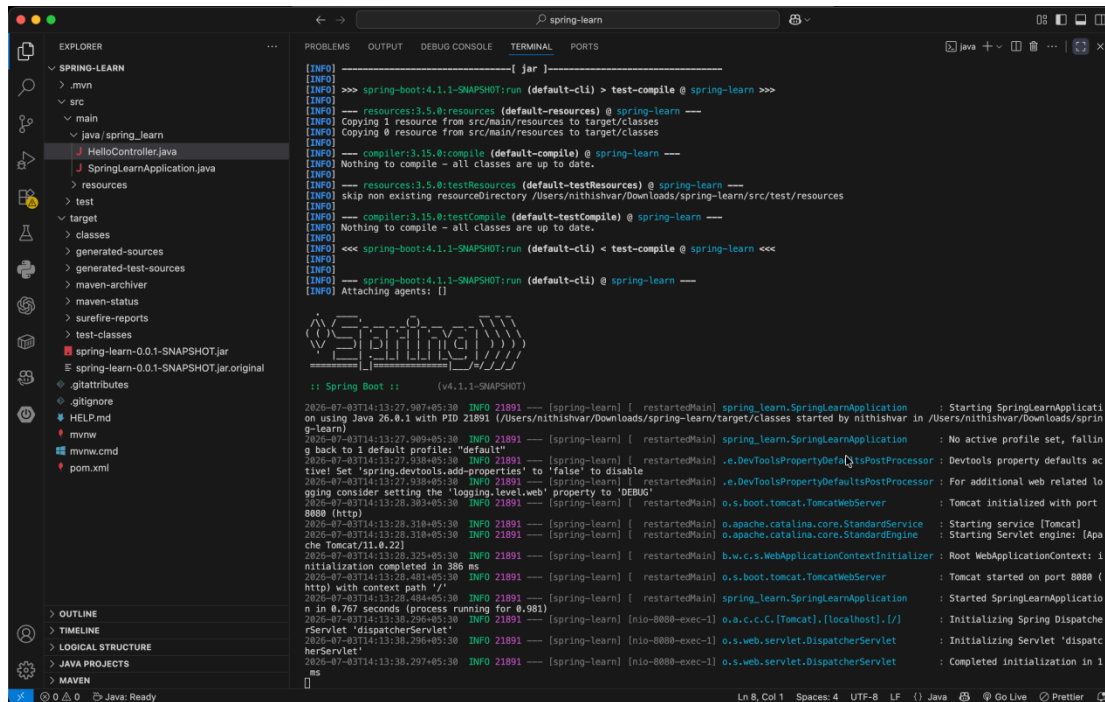
Spring REST using Spring Boot 3







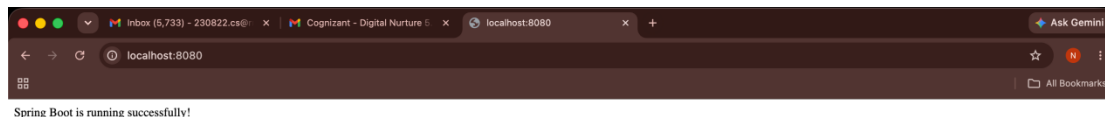
OUTPUT



```
[INFO] -----[ jar ]-----
[INFO]
[INFO] >>> spring-boot:4.1.1-SNAPSHOT:run (default-cli) > test-compile @ spring-learn >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ spring-learn ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ spring-learn ---
[INFO] skip non existing resourceDirectory /Users/nithishvar/Downloads/spring-learn/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:4.1.1-SNAPSHOT:run (default-cli) < test-compile @ spring-learn <<<
[INFO]
[INFO] --- spring-boot:4.1.1-SNAPSHOT:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []

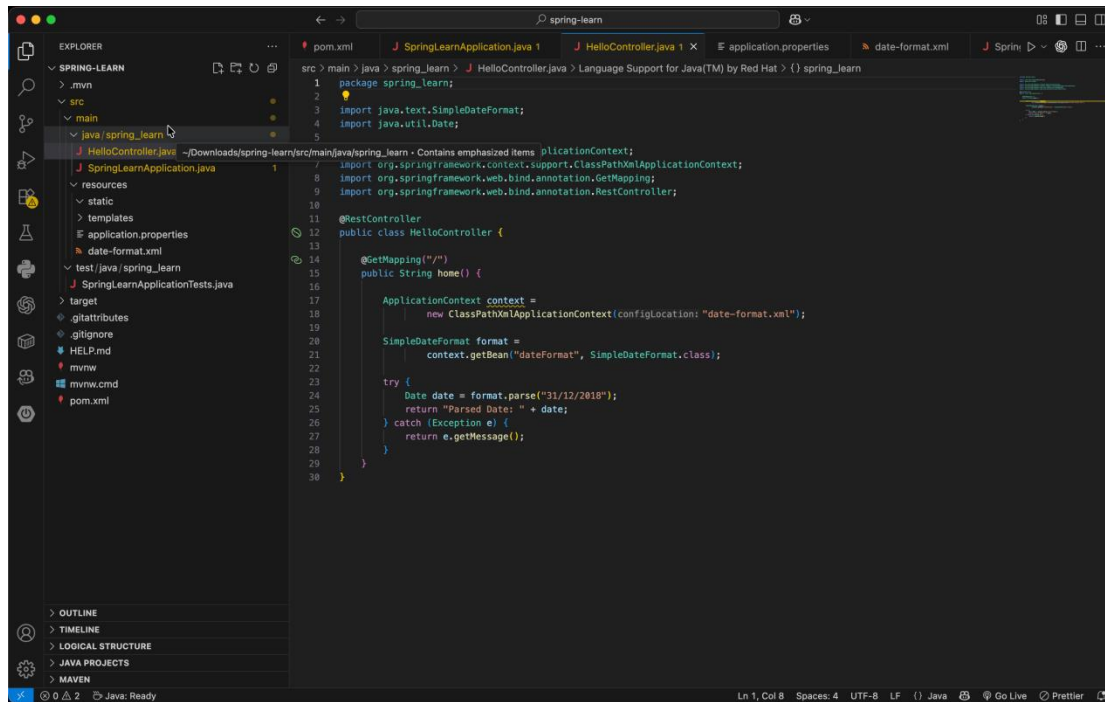
:: Spring Boot ::
(v4.1.1-SNAPSHOT)

2026-07-03T14:13:27.987+05:30 INFO 21891 --- [spring-learn] [ restartedMain] spring.learn.SpringLearnApplication : Starting SpringLearnApplication using Java 26.0.1 with PID 21891 (/Users/nithishvar/Downloads/spring-learn/target/classes started by nithishvar in /Users/nithishvar/Downloads/spring-learn)
2026-07-03T14:13:27.989+05:30 INFO 21891 --- [spring-learn] [ restartedMain] spring.learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-03T14:13:27.939+05:30 INFO 21891 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults are active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-03T14:13:27.939+05:30 INFO 21891 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related loggers, consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-03T14:13:28.383+05:30 INFO 21891 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-03T14:13:28.310+05:30 INFO 21891 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-03T14:13:28.310+05:30 INFO 21891 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-03T14:13:28.325+05:30 INFO 21891 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 386 ms
2026-07-03T14:13:28.484+05:30 INFO 21891 --- [spring-learn] [ restartedMain] spring.learn.SpringLearnApplication : Started SpringLearnApplication in 0.767 seconds (process running for 0.881)
2026-07-03T14:13:38.296+05:30 INFO 21891 --- [spring-learn] [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-03T14:13:38.296+05:30 INFO 21891 --- [spring-learn] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-03T14:13:38.297+05:30 INFO 21891 --- [spring-learn] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
```

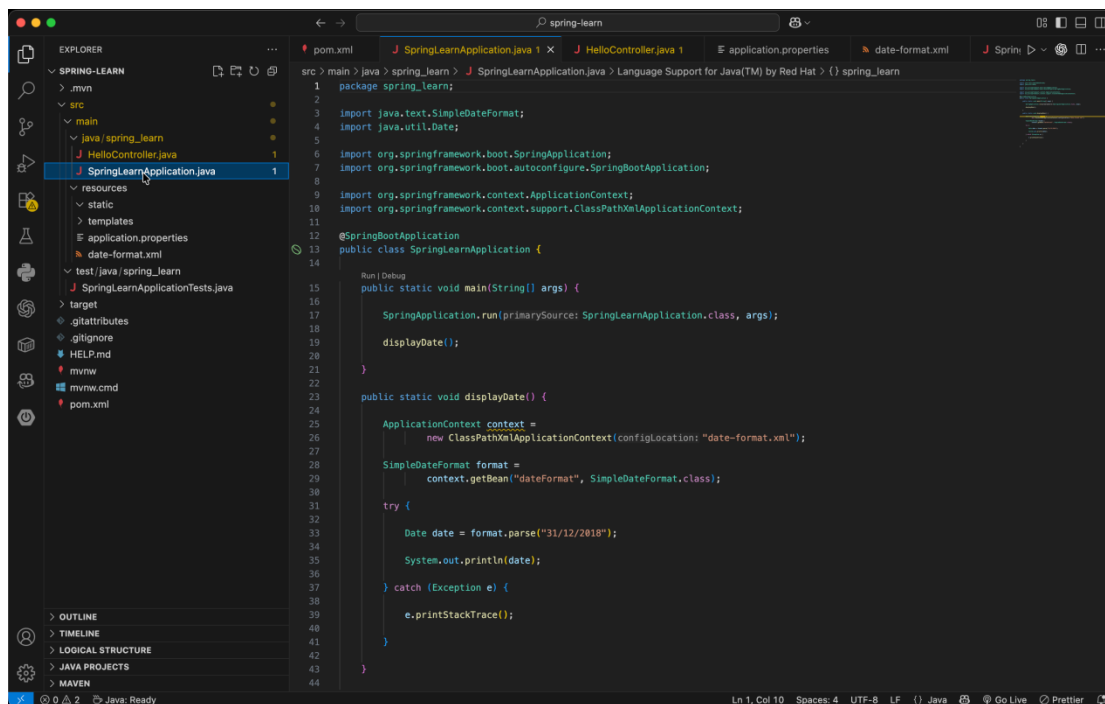


Spring Boot is running successfully!

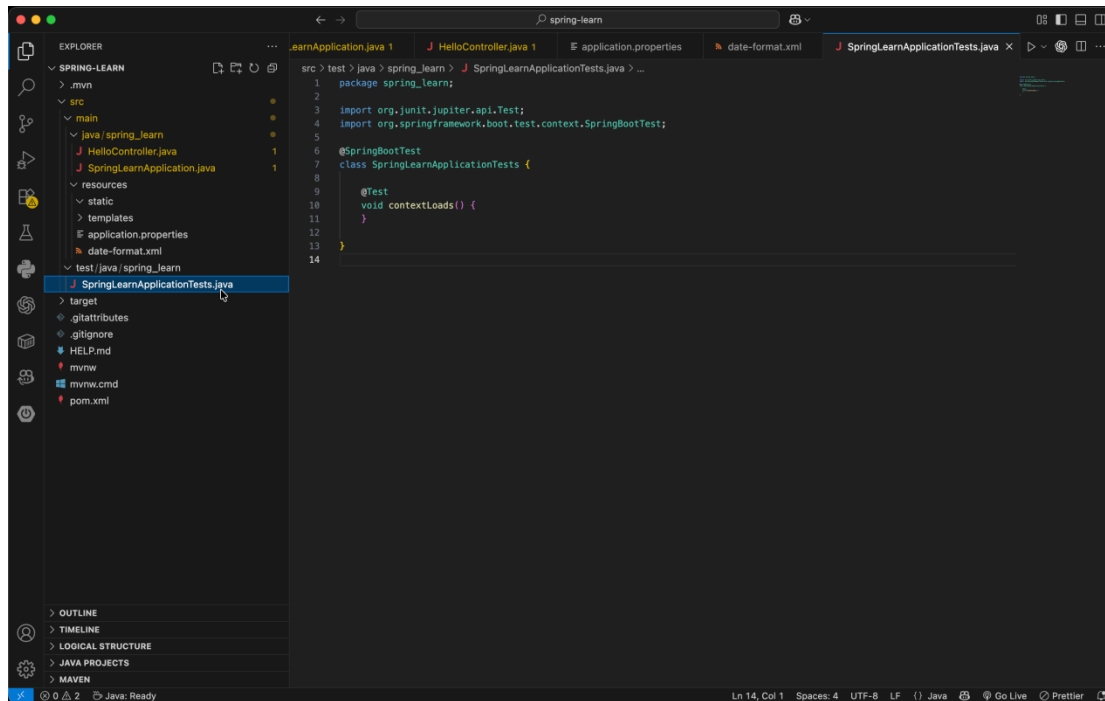
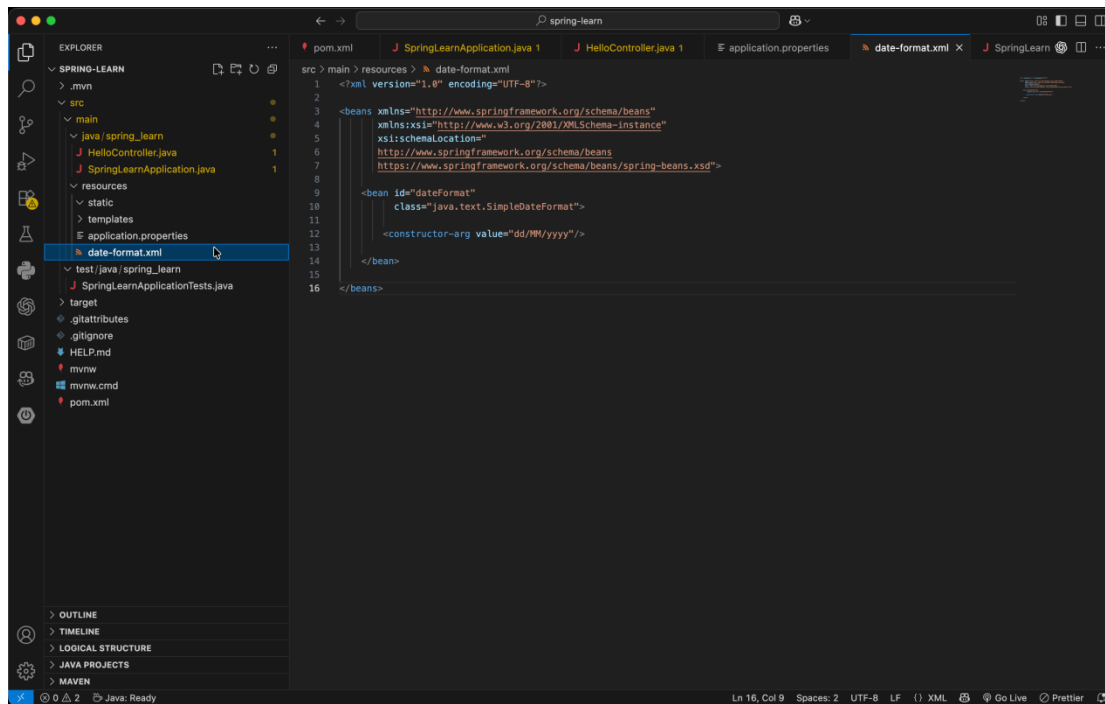
Spring Core – Load Country from Spring Configuration XML

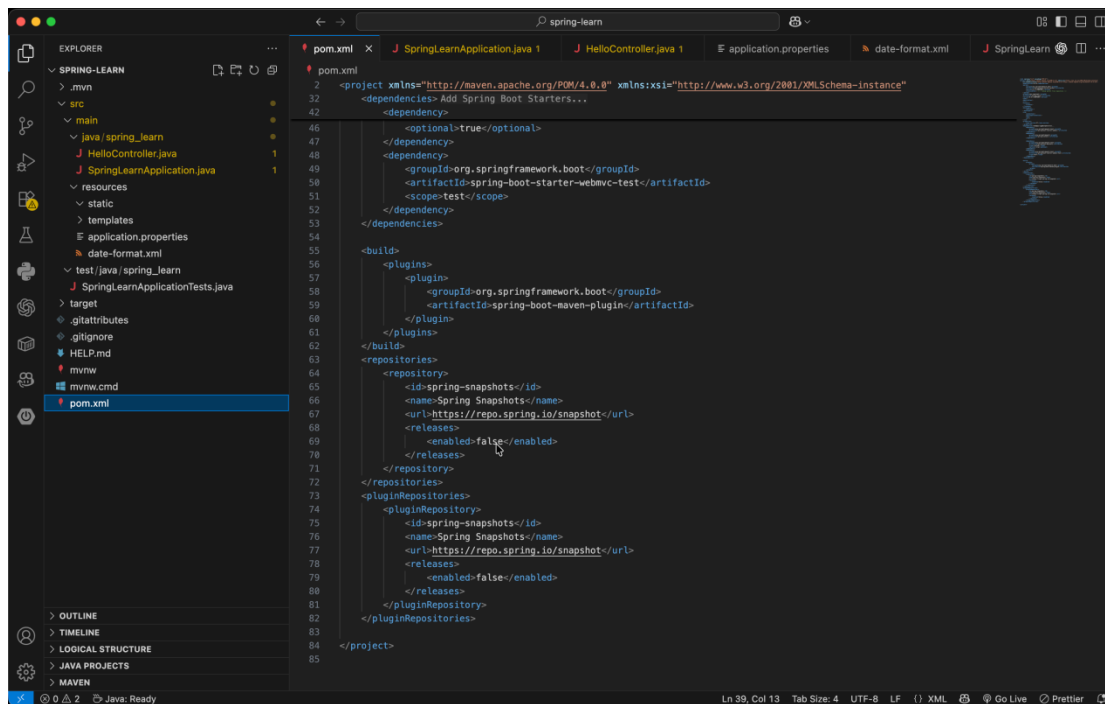
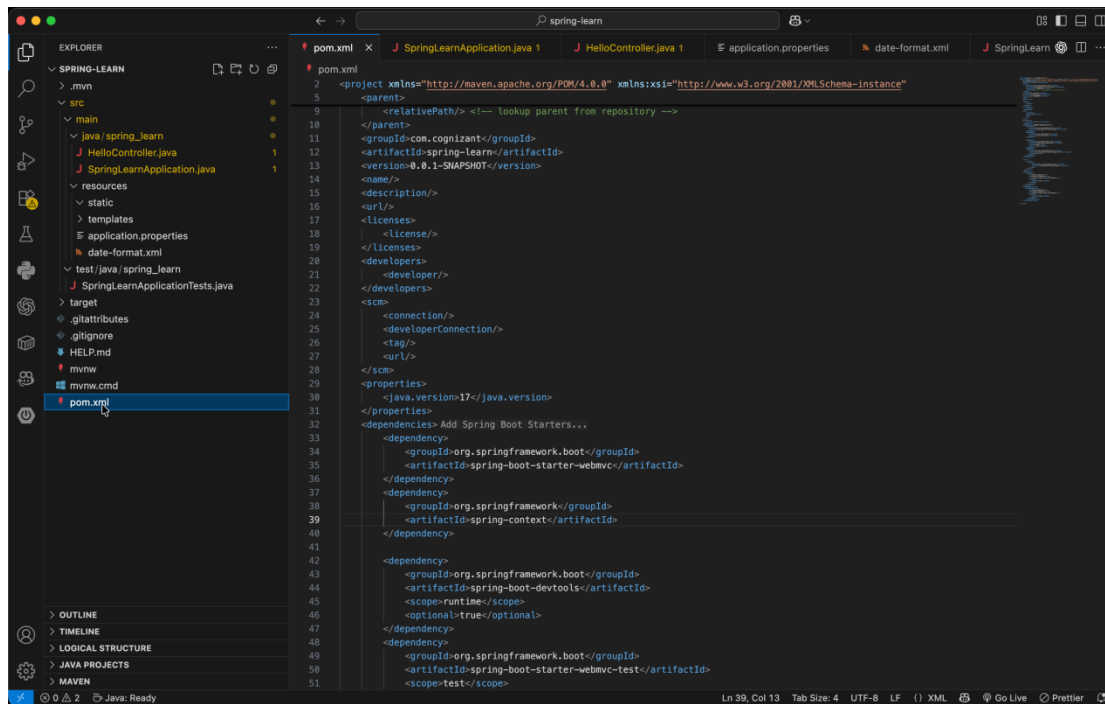


```
1 package spring_learn;
2
3 import java.text.SimpleDateFormat;
4 import java.util.Date;
5
6 import org.springframework.context.support.ClassPathXmlApplicationContext;
7 import org.springframework.web.bind.annotation.GetMapping;
8 import org.springframework.web.bind.annotation.RestController;
9
10 @RestController
11 public class HelloController {
12
13     @GetMapping("/")
14     public String home() {
15
16         ApplicationContext context =
17             new ClassPathXmlApplicationContext("date-format.xml");
18
19         SimpleDateFormat format =
20             context.getBean("dateFormat", SimpleDateFormat.class);
21
22         try {
23             Date date = format.parse("31/12/2018");
24             return "Parsed Date: " + date;
25         } catch (Exception e) {
26             return e.getMessage();
27         }
28     }
29 }
30
```



```
1 package spring_learn;
2
3 import java.text.SimpleDateFormat;
4 import java.util.Date;
5
6 import org.springframework.boot.SpringApplication;
7 import org.springframework.boot.autoconfigure.SpringBootApplication;
8
9 import org.springframework.context.ApplicationContext;
10 import org.springframework.context.support.ClassPathXmlApplicationContext;
11
12 @SpringBootApplication
13 public class SpringLearnApplication {
14
15     Run [Debug]
16     public static void main(String[] args) {
17
18         SpringApplication.run(primarySource: SpringLearnApplication.class, args);
19
20         displayDate();
21     }
22
23     public static void displayDate() {
24
25         ApplicationContext context =
26             new ClassPathXmlApplicationContext("date-format.xml");
27
28         SimpleDateFormat format =
29             context.getBean("dateFormat", SimpleDateFormat.class);
30
31         try {
32             Date date = format.parse("31/12/2018");
33             System.out.println(date);
34         } catch (Exception e) {
35             e.printStackTrace();
36         }
37     }
38 }
39
40
41
42
43
44
```





OUTPUT

The screenshot shows an IDE window titled 'spring-learn'. The left sidebar contains an 'EXPLORER' panel with a file tree for 'SPRING-LEARN'. The main area is split into 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'OUTPUT' tab is active, displaying logs for 'Spring Boot' (v4.1.1-SNAPSHOT). The logs show the application starting on port 8081, with various components like Tomcat, Spring, and the application itself. The status bar at the bottom indicates 'Ln 1, Col 8, Spaces: 4, UTF-8, LF, Java, Go Live, Prettier'.

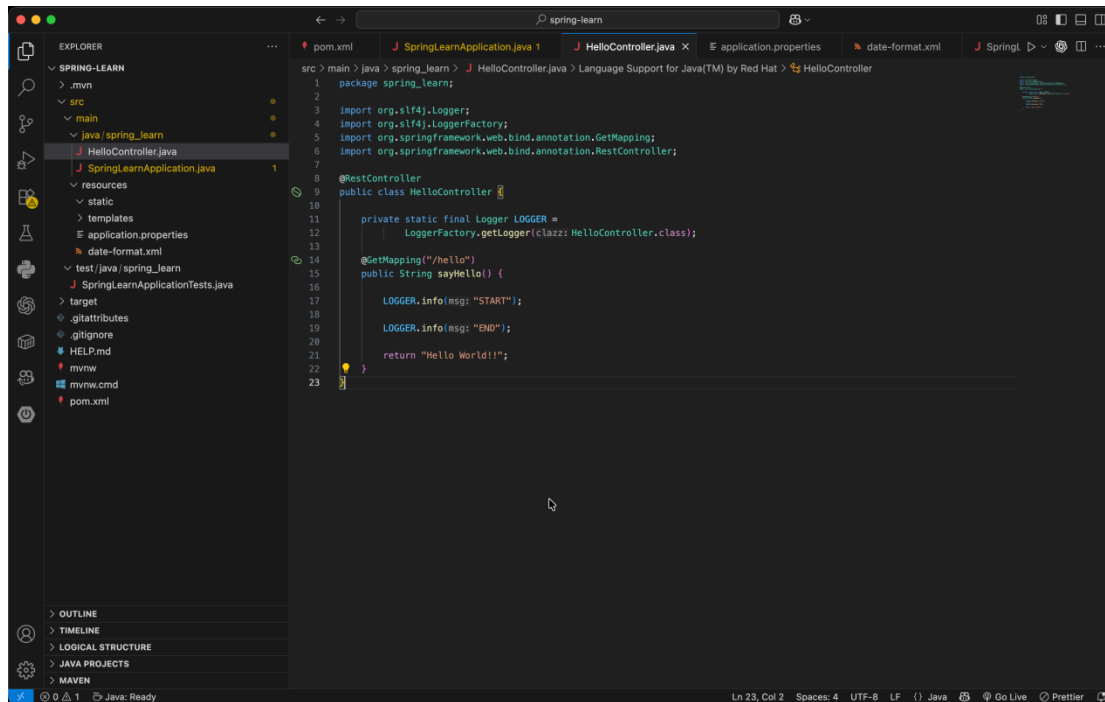
```
2026-07-04T09:29:47.838+05:30 INFO 33637 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Starting Spr
ingLearnApplication using Java 26.0.1 with PID 33637 (/Users/nithishvar/Downloads/spring-learn/target/classes started by nithishvar in
/Users/nithishvar/Downloads/spring-learn)
2026-07-04T09:29:47.838+05:30 INFO 33637 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : No active pr
ofile set, falling back to 1 default profile: "default"
2026-07-04T09:29:47.111+05:30 INFO 33637 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initi
alized with port 8081 (http)
2026-07-04T09:29:47.112+05:30 INFO 33637 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting ser
vice [Tomcat]
2026-07-04T09:29:47.112+05:30 INFO 33637 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Ser
vice [Apache Tomcat/11.0.22]
2026-07-04T09:29:47.116+05:30 INFO 33637 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebAppl
icationContext: initialization completed in 77 ms
2026-07-04T09:29:47.152+05:30 INFO 33637 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat start
ed on port 8081 (http) with context path '/'
2026-07-04T09:29:47.154+05:30 INFO 33637 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Started Spr
ingLearnApplication in 0.129 seconds (process running for 52.063)
2026-07-04T09:29:47.155+05:30 INFO 33637 --- [spring-learn] [ restartedMain] .ConditionEvaluationDeltaLoggingListener : Condition ev
aluation unchanged
Mon Dec 31 00:00:00 IST 2018
2026-07-04T09:29:51.588+05:30 INFO 33637 --- [spring-learn] [ File Watcher] rtingClassPathChangeEventListeners : Restarting d
ue to 1 class path change (1 addition, 0 deletions, 0 modifications)
2026-07-04T09:29:51.591+05:30 INFO 33637 --- [spring-learn] [ Thread-5] o.s.boot.tomcat.GracefulShutdown : Commencing g
raceful shutdown. Waiting for active requests to complete
2026-07-04T09:29:51.593+05:30 INFO 33637 --- [spring-learn] [ tomcat-shutdown] o.s.boot.tomcat.GracefulShutdown : Graceful shu
tdown complete

Spring
=====

2026-07-04T09:29:51.634+05:30 INFO 33637 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Starting Spr
ingLearnApplication using Java 26.0.1 with PID 33637 (/Users/nithishvar/Downloads/spring-learn/target/classes started by nithishvar in
/Users/nithishvar/Downloads/spring-learn)
2026-07-04T09:29:51.634+05:30 INFO 33637 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : No active pr
ofile set, falling back to 1 default profile: "default"
2026-07-04T09:29:51.692+05:30 INFO 33637 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initi
alized with port 8081 (http)
2026-07-04T09:29:51.692+05:30 INFO 33637 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting ser
vice [Tomcat]
2026-07-04T09:29:51.692+05:30 INFO 33637 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Ser
vice [Apache Tomcat/11.0.22]
2026-07-04T09:29:51.699+05:30 INFO 33637 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebAppl
icationContext: initialization completed in 60 ms
2026-07-04T09:29:51.725+05:30 INFO 33637 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat start
ed on port 8081 (http) with context path '/'
2026-07-04T09:29:51.726+05:30 INFO 33637 --- [spring-learn] [ restartedMain] c.c.springlearn.SpringLearnApplication : Started Spr
ingLearnApplication in 0.181 seconds (process running for 56.634)
2026-07-04T09:29:51.727+05:30 INFO 33637 --- [spring-learn] [ restartedMain] .ConditionEvaluationDeltaLoggingListener : Condition ev
aluation unchanged
Mon Dec 31 00:00:00 IST 2018
```

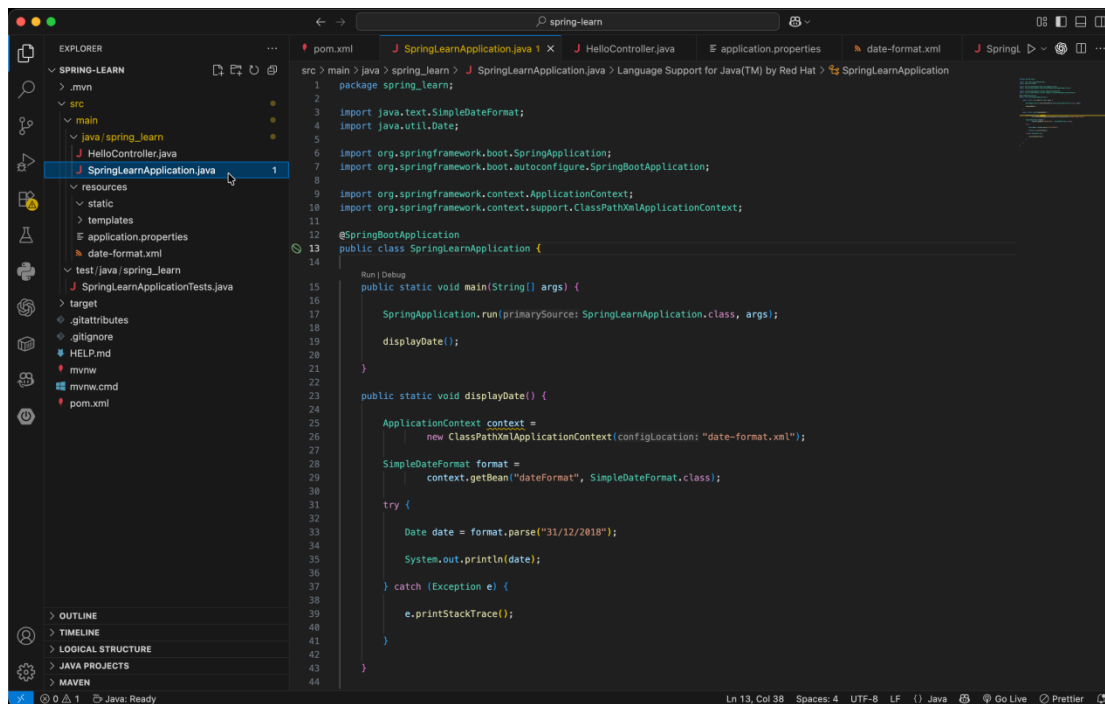


Hello World RESTful Web Service



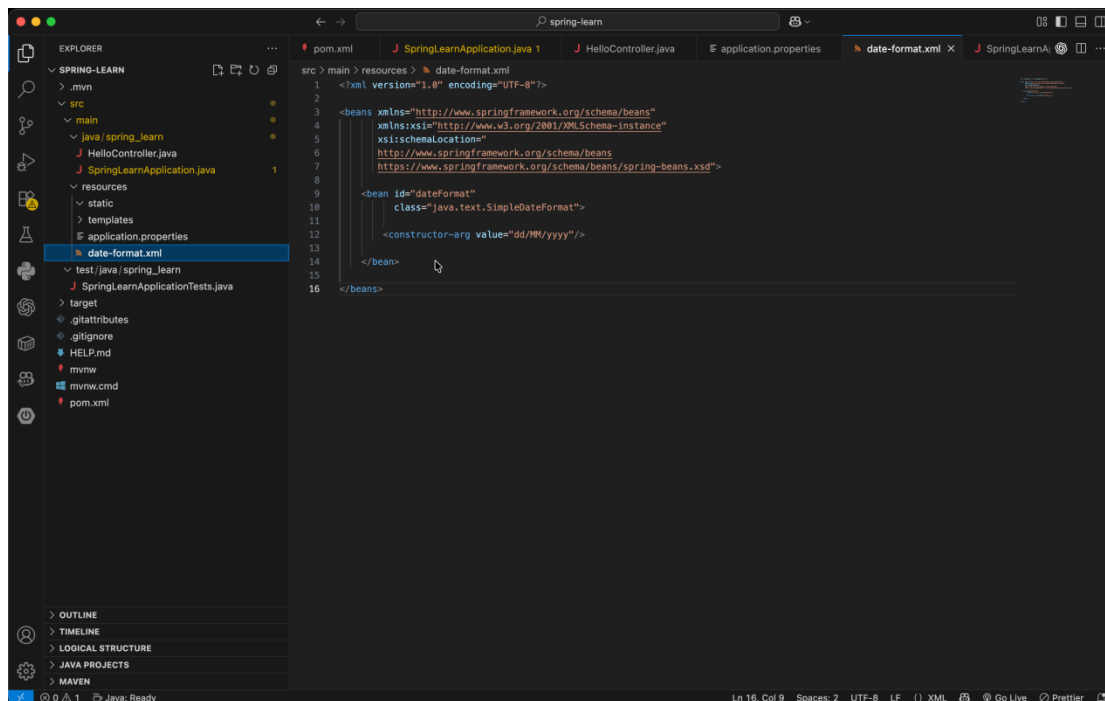
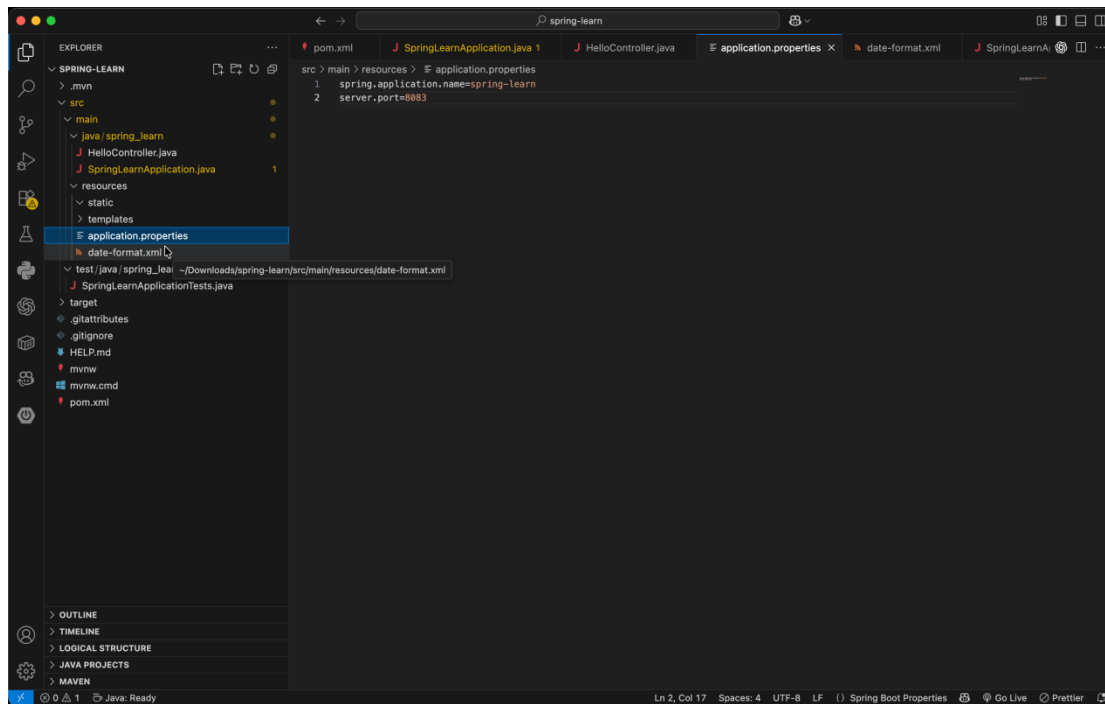
This screenshot shows an IDE window for a project named 'spring-learn'. The Explorer panel on the left shows the project structure, with 'HelloController.java' selected under 'src/main/java/spring_learn'. The main editor displays the code for 'HelloController.java', which is a REST controller. The code includes imports for logging and Spring's REST framework, and implements a 'sayHello()' method that returns 'Hello World!!'.

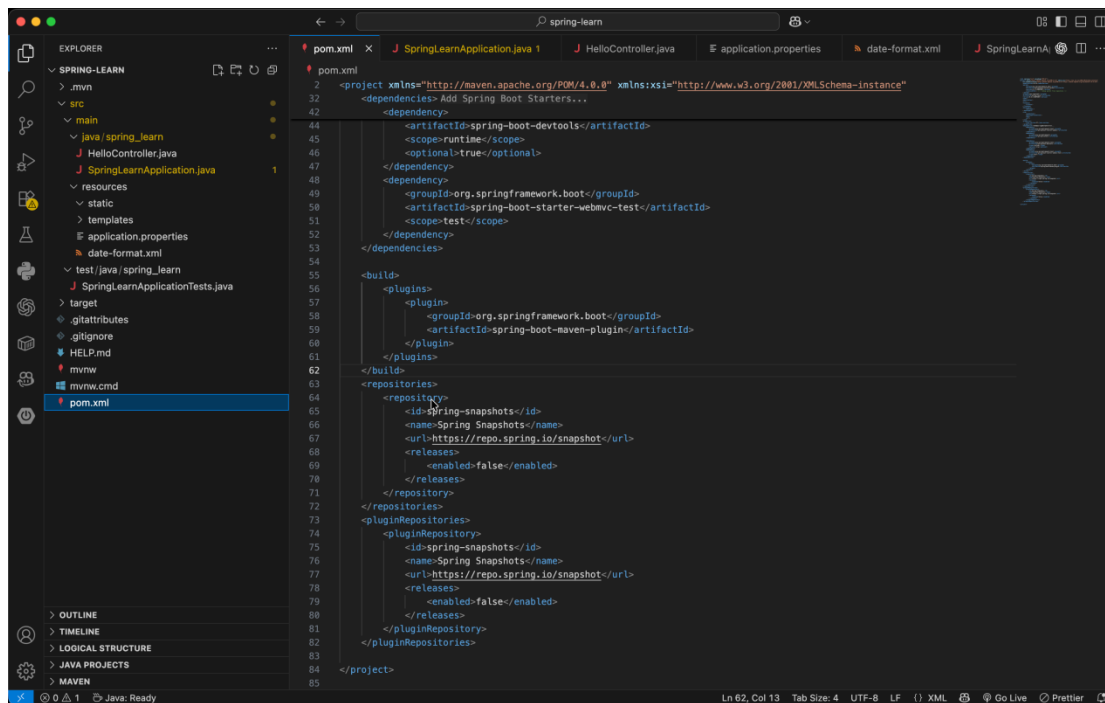
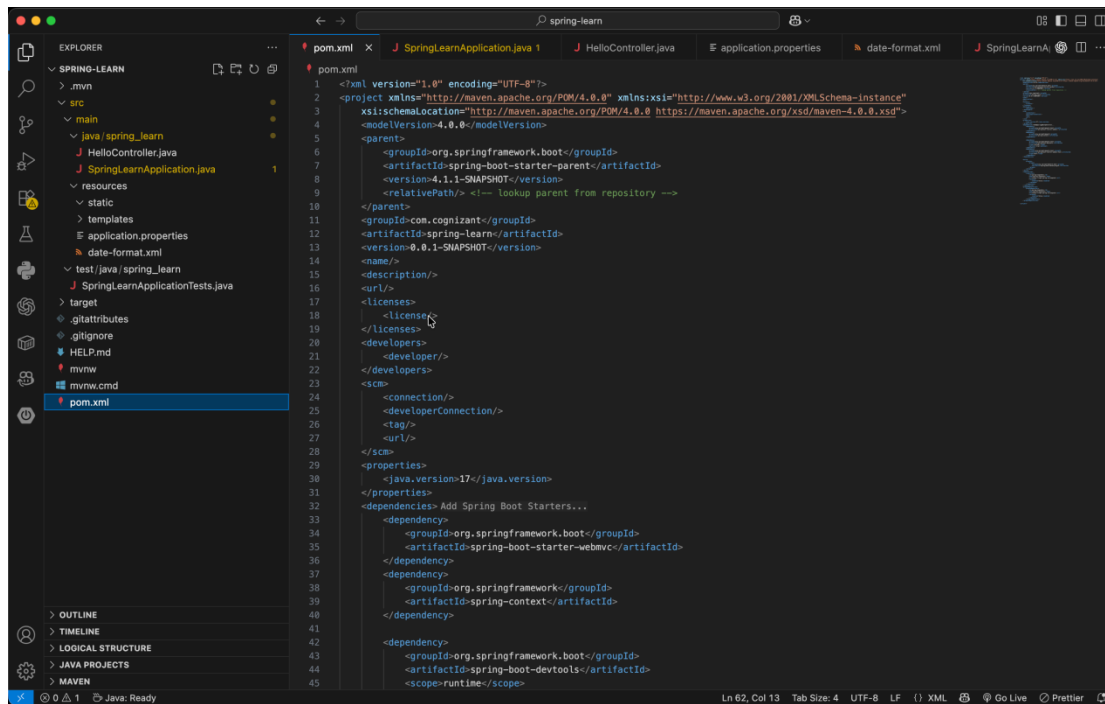
```
src > main > java > spring_learn > J HelloController.java > Language Support for Java(TM) by Red Hat > HelloController
1 package spring_learn;
2
3 import org.slf4j.Logger;
4 import org.slf4j.LoggerFactory;
5 import org.springframework.web.bind.annotation.GetMapping;
6 import org.springframework.web.bind.annotation.RestController;
7
8 @RestController
9 public class HelloController {
10
11     private static final Logger LOGGER =
12         LoggerFactory.getLogger(clazz: HelloController.class);
13
14     @GetMapping("/hello")
15     public String sayHello() {
16
17         LOGGER.info(msg: "START");
18
19         LOGGER.info(msg: "END");
20
21         return "Hello World!!";
22     }
23 }
```



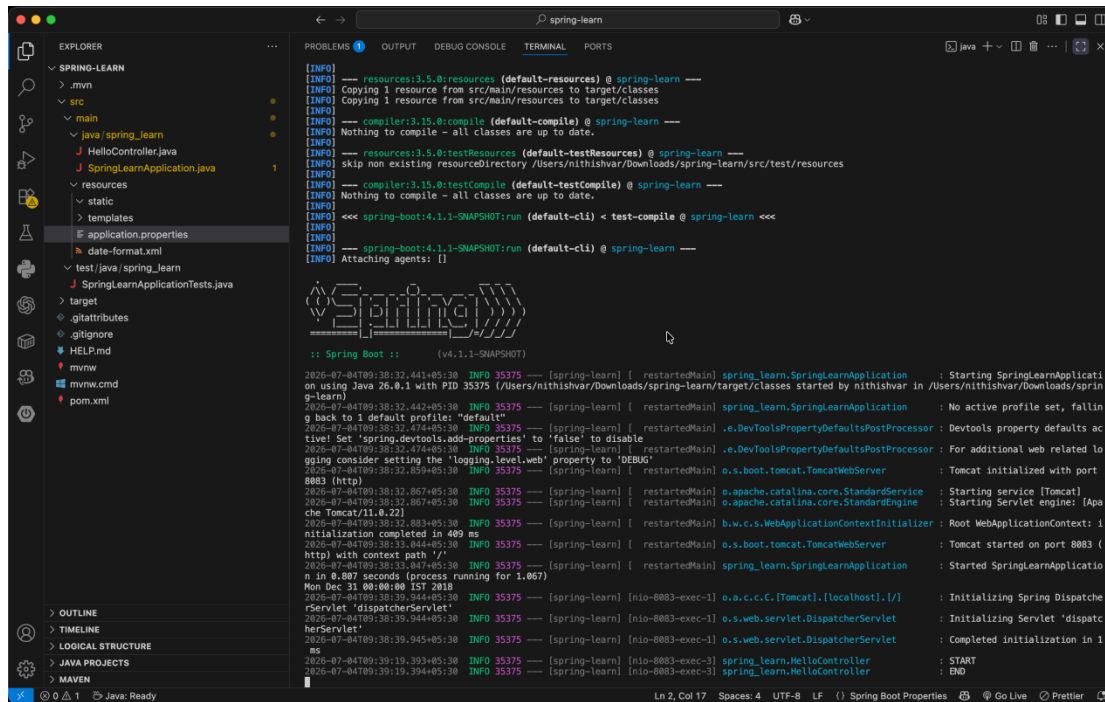
This screenshot shows the same IDE window, but now 'SpringLearnApplication.java' is selected in the Explorer panel. The main editor displays the code for 'SpringLearnApplication.java', which is a Spring Boot application. The code includes imports for Spring's application and context frameworks, and implements a 'main()' method that runs the application and a 'displayDate()' method that parses a date string.

```
src > main > java > spring_learn > J SpringLearnApplication.java > Language Support for Java(TM) by Red Hat > SpringLearnApplication
1 package spring_learn;
2
3 import java.text.SimpleDateFormat;
4 import java.util.Date;
5
6 import org.springframework.boot.SpringApplication;
7 import org.springframework.boot.autoconfigure.SpringBootApplication;
8
9 import org.springframework.context.ApplicationContext;
10 import org.springframework.context.support.ClassPathXmlApplicationContext;
11
12 @SpringBootApplication
13 public class SpringLearnApplication {
14
15     Run | Debug
16     public static void main(String[] args) {
17
18         SpringApplication.run(primarySource: SpringLearnApplication.class, args);
19
20         displayDate();
21     }
22
23     public static void displayDate() {
24
25         ApplicationContext context =
26             new ClassPathXmlApplicationContext(configLocation: "date-format.xml");
27
28         SimpleDateFormat format =
29             context.getBean("dateFormat", SimpleDateFormat.class);
30
31         try {
32
33             Date date = format.parse("31/12/2018");
34             System.out.println(date);
35
36         } catch (Exception e) {
37
38             e.printStackTrace();
39         }
40     }
41
42 }
43
44 }
```





OUTPUT



```
[INFO] --- resources:3.5.0:resources (default-resources) @ spring-learn ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] --- compiler:3.15.0:compile (default-compile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.5.0:testResources (default-testResources) @ spring-learn ---
[INFO] skip non existing resourceDirectory /Users/nithishvar/Downloads/spring-learn/src/test/resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] <<< spring-boot:4.1.1-SNAPSHOT:run (default-cli) < test-compile @ spring-learn <<<
[INFO] --- spring-boot:4.1.1-SNAPSHOT:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []

Spring
=====
:: Spring Boot :: (v4.1.1-SNAPSHOT)

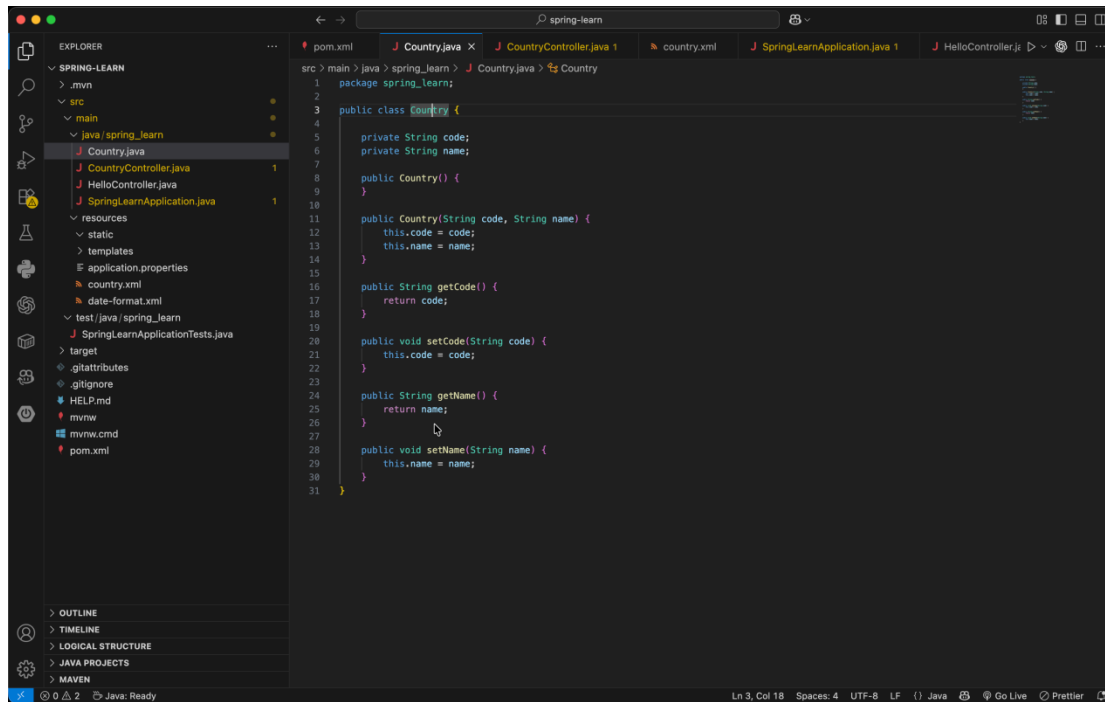
2026-07-04T09:38:32.441+05:30 INFO 35375 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Starting SpringLearnApplication on using Java 26.0.1 with PID 35375 (/Users/nithishvar/Downloads/spring-learn/target/classes started by nithishvar in /Users/nithishvar/Downloads/spring-learn)
2026-07-04T09:38:32.442+05:30 INFO 35375 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-04T09:38:32.474+05:30 INFO 35375 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults are active! Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-04T09:38:32.474+05:30 INFO 35375 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-04T09:38:32.859+05:30 INFO 35375 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-04T09:38:32.867+05:30 INFO 35375 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-04T09:38:32.867+05:30 INFO 35375 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-04T09:38:32.883+05:30 INFO 35375 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 409 ms
2026-07-04T09:38:33.184+05:30 INFO 35375 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-04T09:38:33.047+05:30 INFO 35375 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Started SpringLearnApplication in 0.887 seconds (process running for 1.067)
2026-07-04T09:38:33.047+05:30 INFO 35375 --- [spring-learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-04T09:38:33.044+05:30 INFO 35375 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-04T09:38:39.945+05:30 INFO 35375 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-07-04T09:39:19.393+05:30 INFO 35375 --- [spring-learn] [nio-8083-exec-3] spring_learn.HelloController : START
2026-07-04T09:39:19.394+05:30 INFO 35375 --- [spring-learn] [nio-8083-exec-3] spring_learn.HelloController : END
```



```
localhost:8083/hello

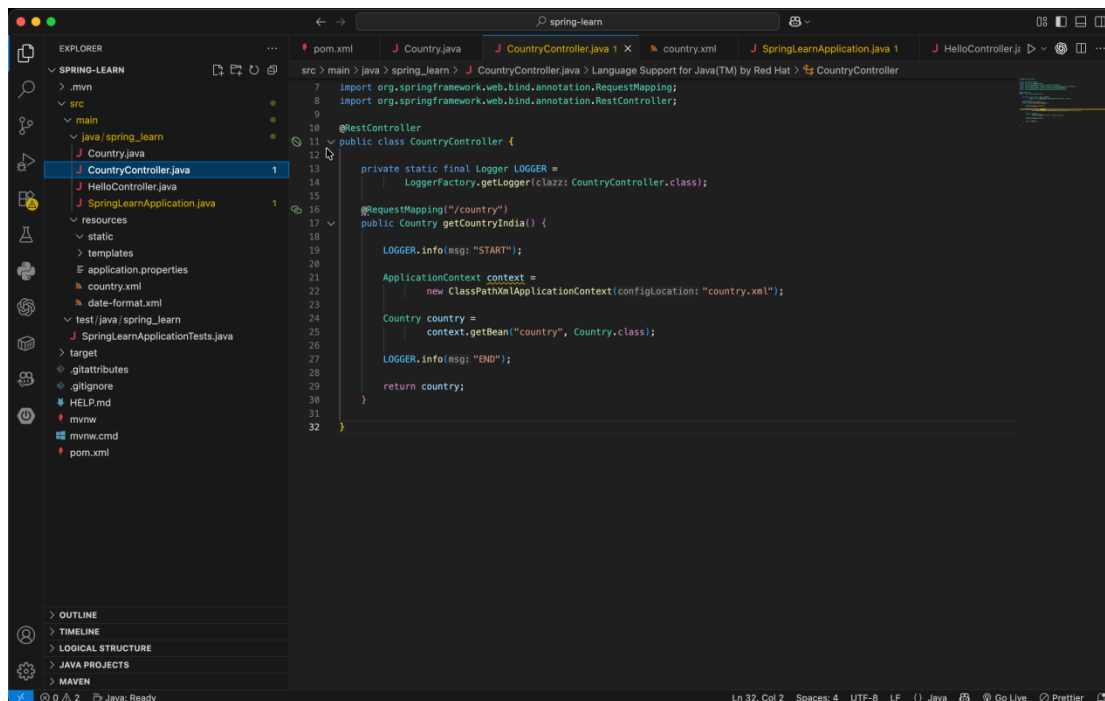
Hello World!!
```


REST - Country Web Service



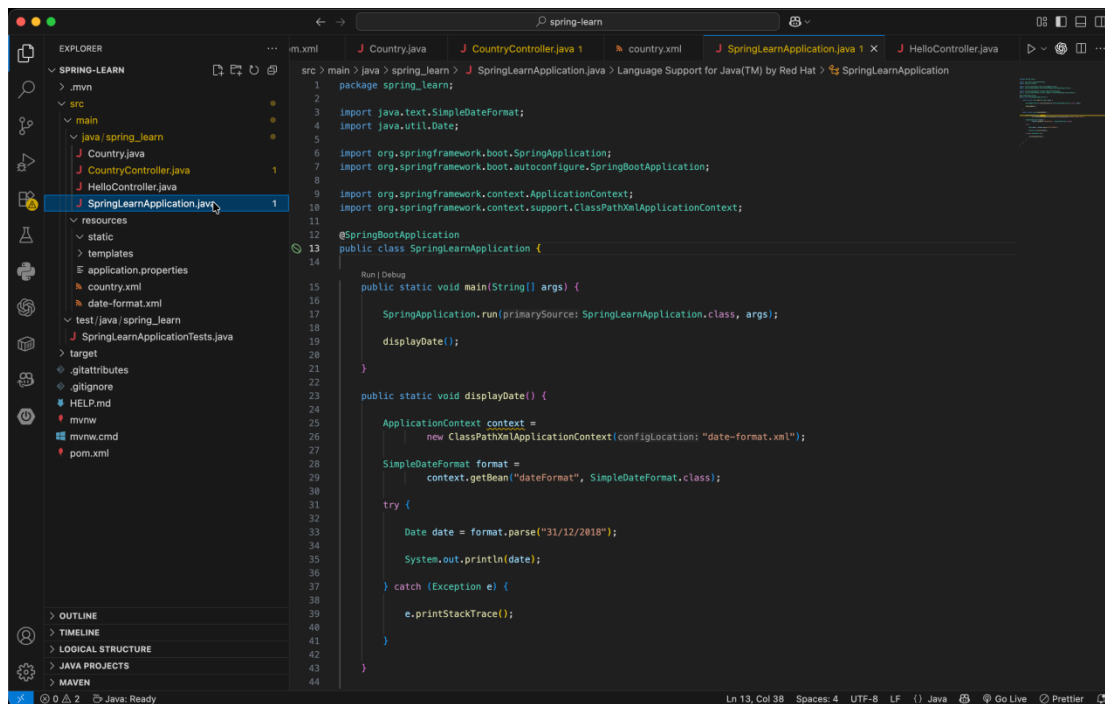
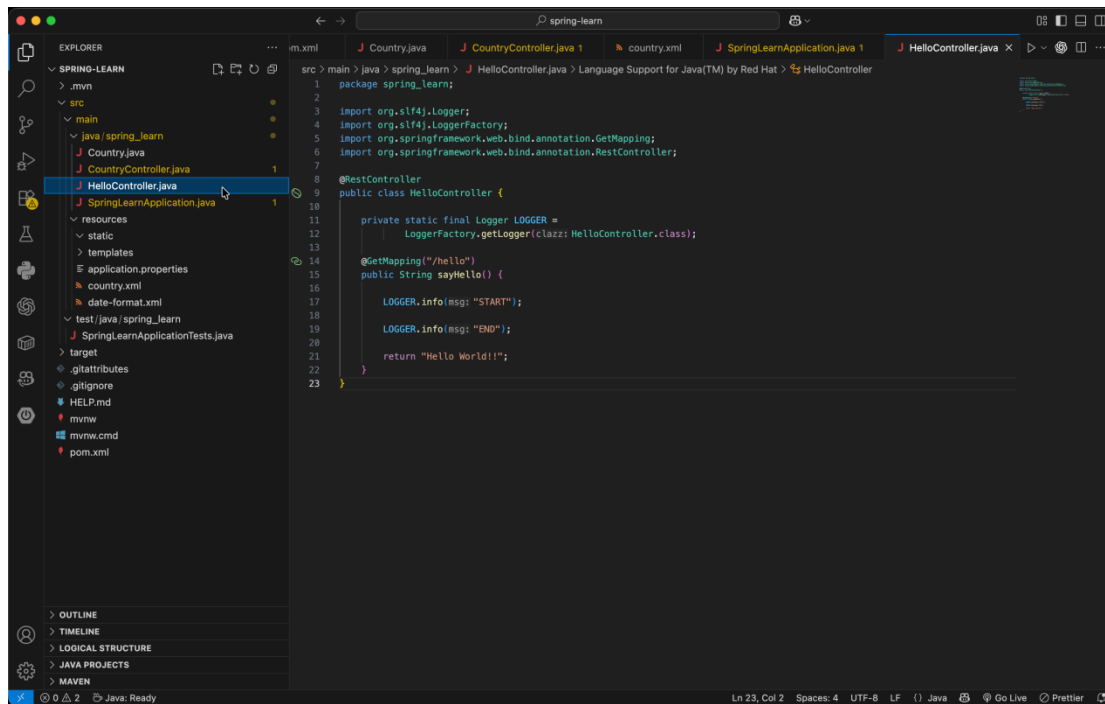
This screenshot shows an IDE with the 'Country.java' file open. The Explorer on the left shows the project structure for 'spring-learn'. The main editor displays the following code:

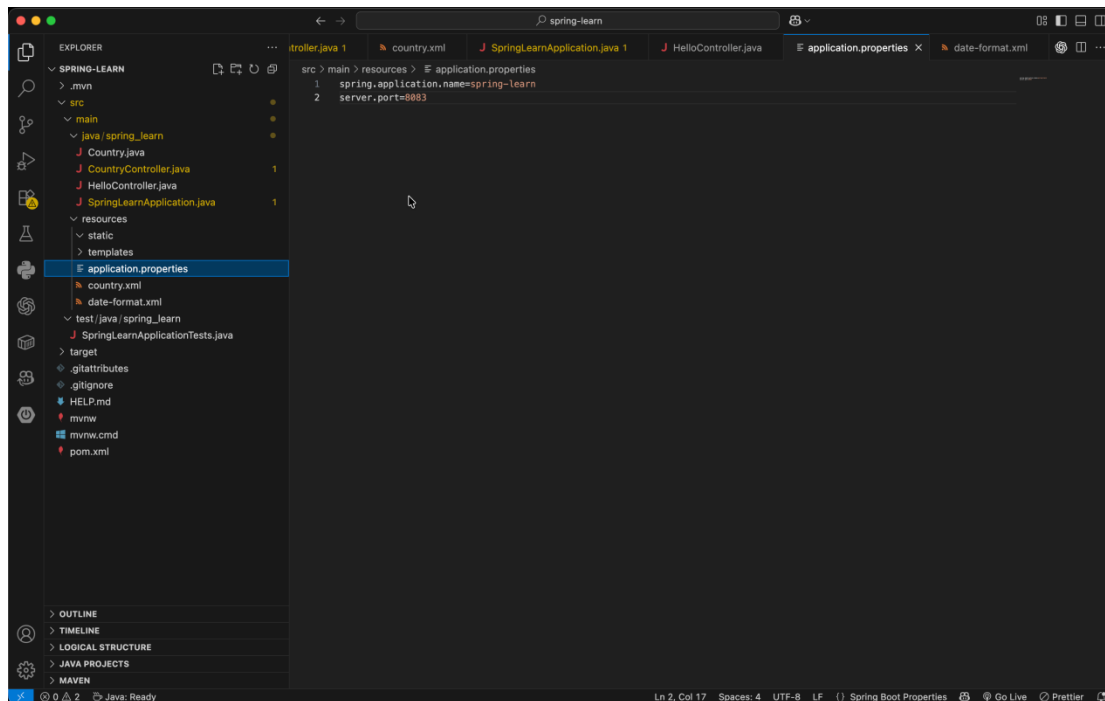
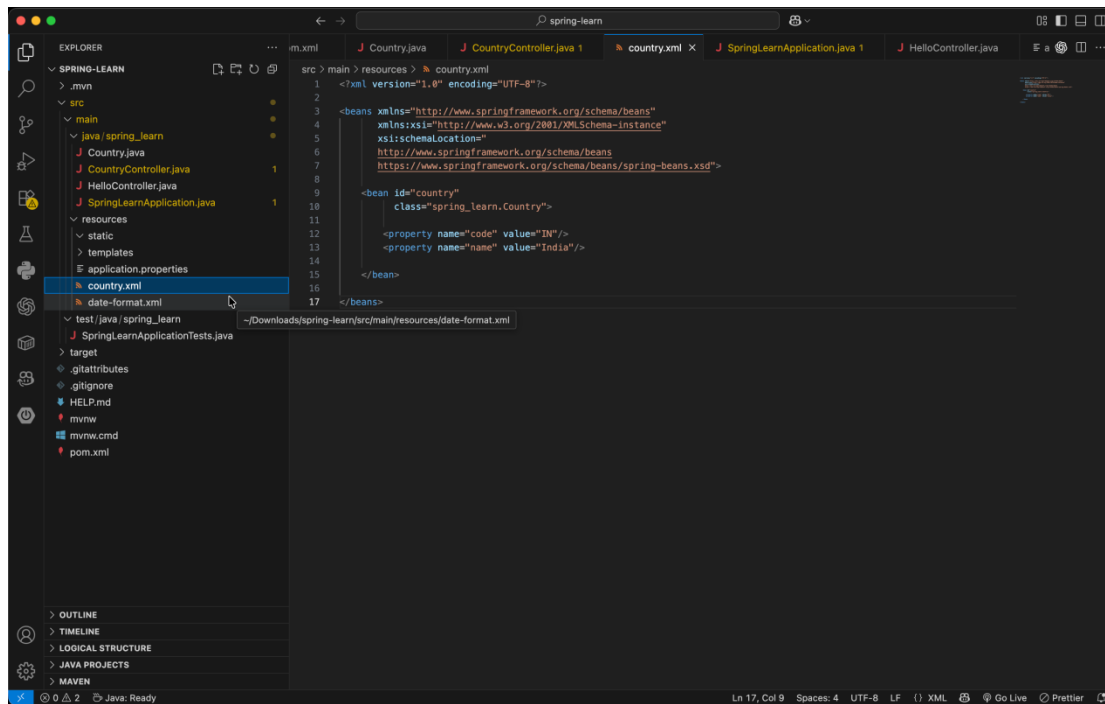
```
1 package spring_learn;
2
3 public class Country {
4     private String code;
5     private String name;
6
7     public Country() {
8     }
9
10    public Country(String code, String name) {
11        this.code = code;
12        this.name = name;
13    }
14
15    public String getCode() {
16        return code;
17    }
18
19    public void setCode(String code) {
20        this.code = code;
21    }
22
23    public String getName() {
24        return name;
25    }
26
27    public void setName(String name) {
28        this.name = name;
29    }
30
31 }
```



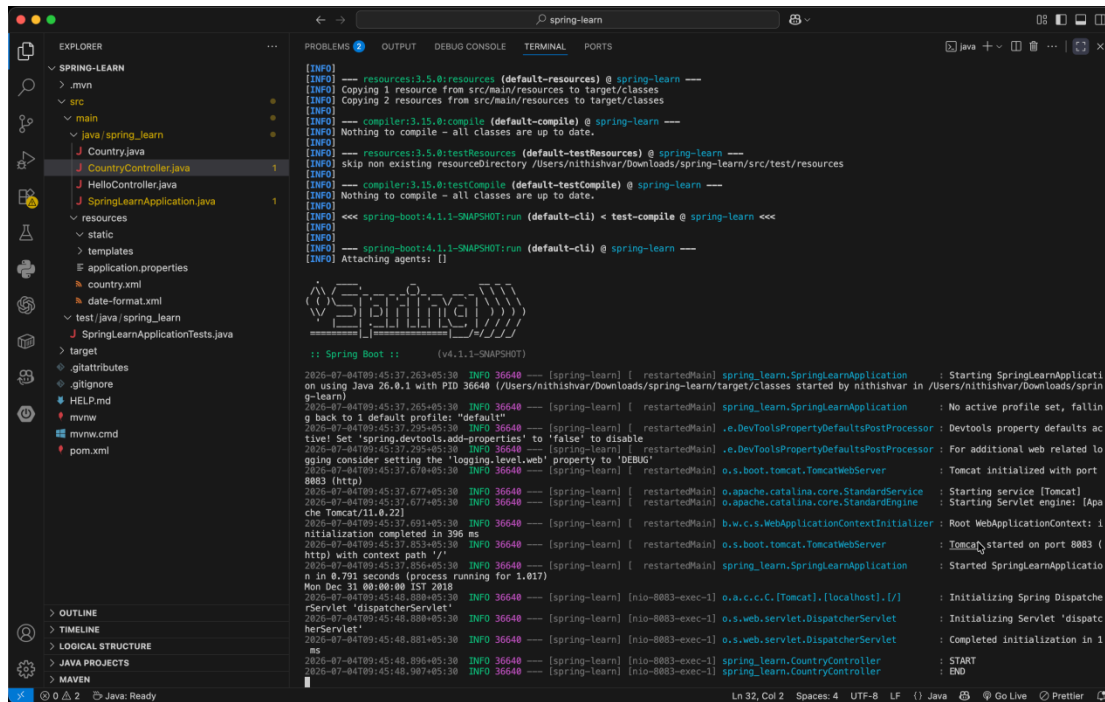
This screenshot shows the same IDE with the 'CountryController.java' file open. The Explorer on the left shows the project structure. The main editor displays the following code:

```
7 import org.springframework.web.bind.annotation.RequestMapping;
8 import org.springframework.web.bind.annotation.RestController;
9
10 @RestController
11 public class CountryController {
12     private static final Logger LOGGER =
13         LoggerFactory.getLogger(clazz: CountryController.class);
14
15     @RequestMapping("/country")
16     public Country getCountryIndia() {
17         LOGGER.info(msg: "START");
18
19         ApplicationContext context =
20             new ClassPathXmlApplicationContext(configLocation: "country.xml");
21
22         Country country =
23             context.getBean("country", Country.class);
24
25         LOGGER.info(msg: "END");
26
27         return country;
28     }
29
30 }
31
32 }
```





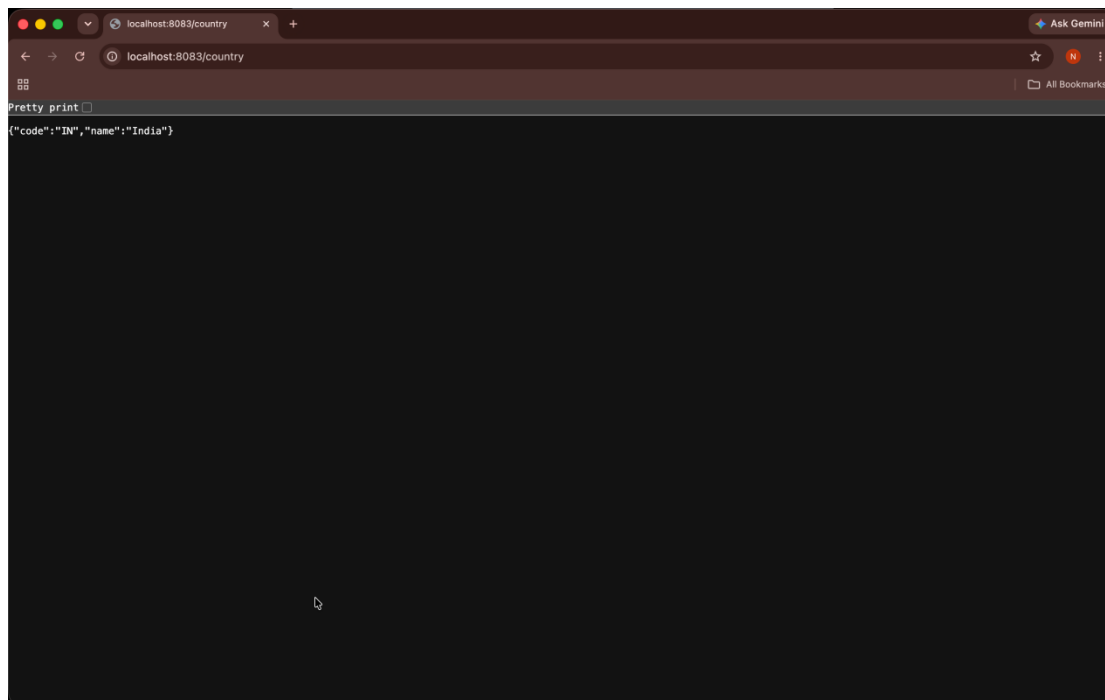
OUTPUT



```
[INFO] --- resources:3.5.0:resources (default-resources) @ spring-learn ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 2 resources from src/main/resources to target/classes
[INFO] --- compiler:3.15.0:compile (default-compile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.5.0:testResources (default-testResources) @ spring-learn ---
[INFO] skip non existing resourceDirectory /Users/nithishvar/Downloads/spring-learn/src/test/resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ spring-learn ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] <<< spring-boot:4.1.1-SNAPSHOT:run (default-cli) < test-compile @ spring-learn <<<
[INFO] --- spring-boot:4.1.1-SNAPSHOT:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []

Spring Boot (v4.1.1-SNAPSHOT)

2026-07-04T09:45:37.263+05:30 INFO 36640 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Starting SpringLearnApplication on using Java 26.0.1 with PID 36640 (/Users/nithishvar/Downloads/spring-learn/target/classes started by nithishvar in /Users/nithishvar/Downloads/spring-learn)
2026-07-04T09:45:37.265+05:30 INFO 36640 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-04T09:45:37.295+05:30 INFO 36640 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults are active. For additional web related logging consider setting the 'logging.level.web' property to 'DEBUG'
2026-07-04T09:45:37.678+05:30 INFO 36640 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-04T09:45:37.677+05:30 INFO 36640 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-04T09:45:37.677+05:30 INFO 36640 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-04T09:45:37.691+05:30 INFO 36640 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 396 ms
2026-07-04T09:45:37.853+05:30 INFO 36640 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-04T09:45:37.856+05:30 INFO 36640 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Started SpringLearnApplication in 0.791 seconds (process running for 1.017)
2026-07-04T09:45:48.888+05:30 INFO 36640 --- [spring-learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-04T09:45:48.888+05:30 INFO 36640 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-04T09:45:48.881+05:30 INFO 36640 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
2026-07-04T09:45:48.896+05:30 INFO 36640 --- [spring-learn] [nio-8083-exec-1] spring_learn.CountryController : START
2026-07-04T09:45:48.907+05:30 INFO 36640 --- [spring-learn] [nio-8083-exec-1] spring_learn.CountryController : END
```

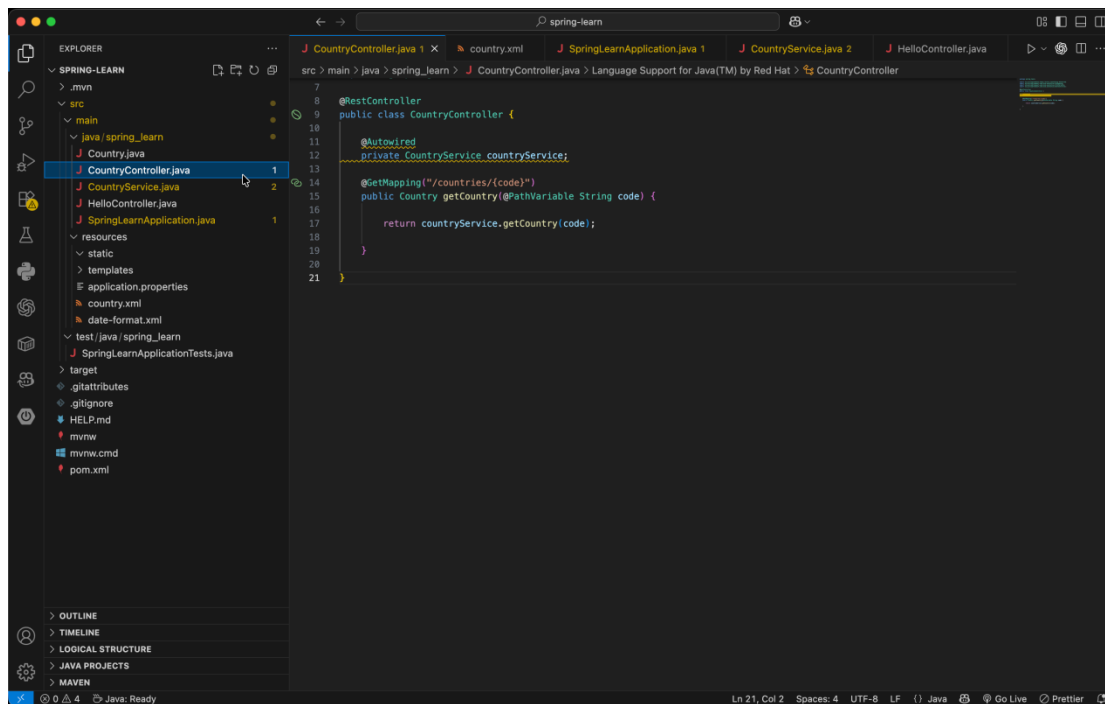
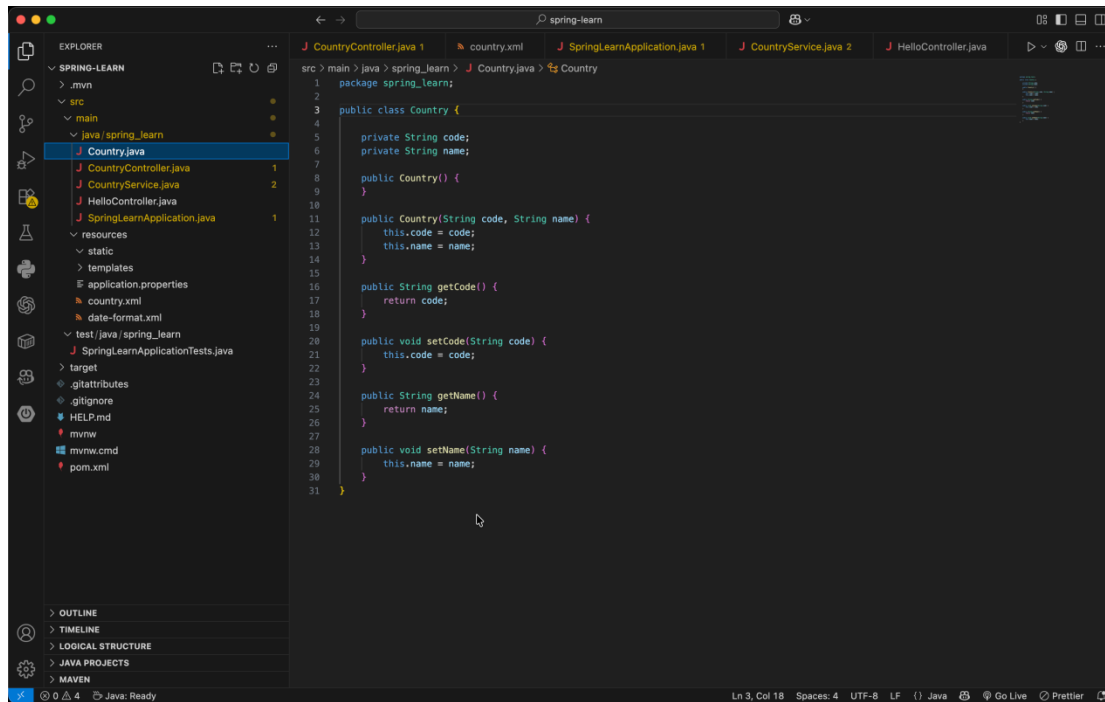


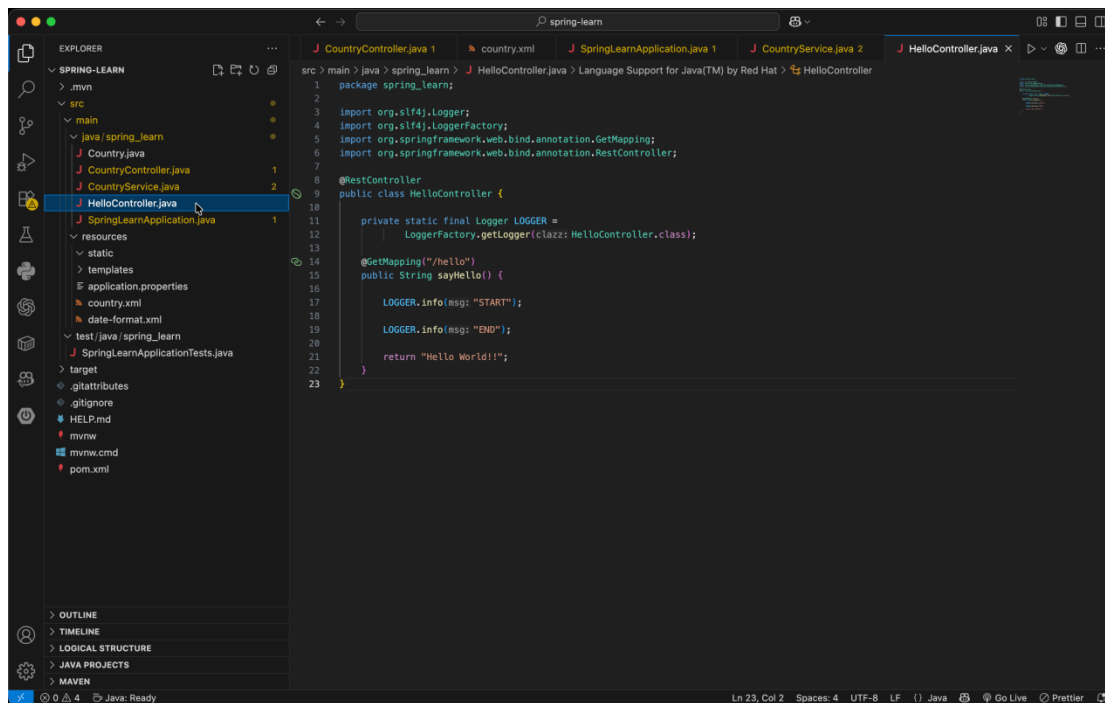
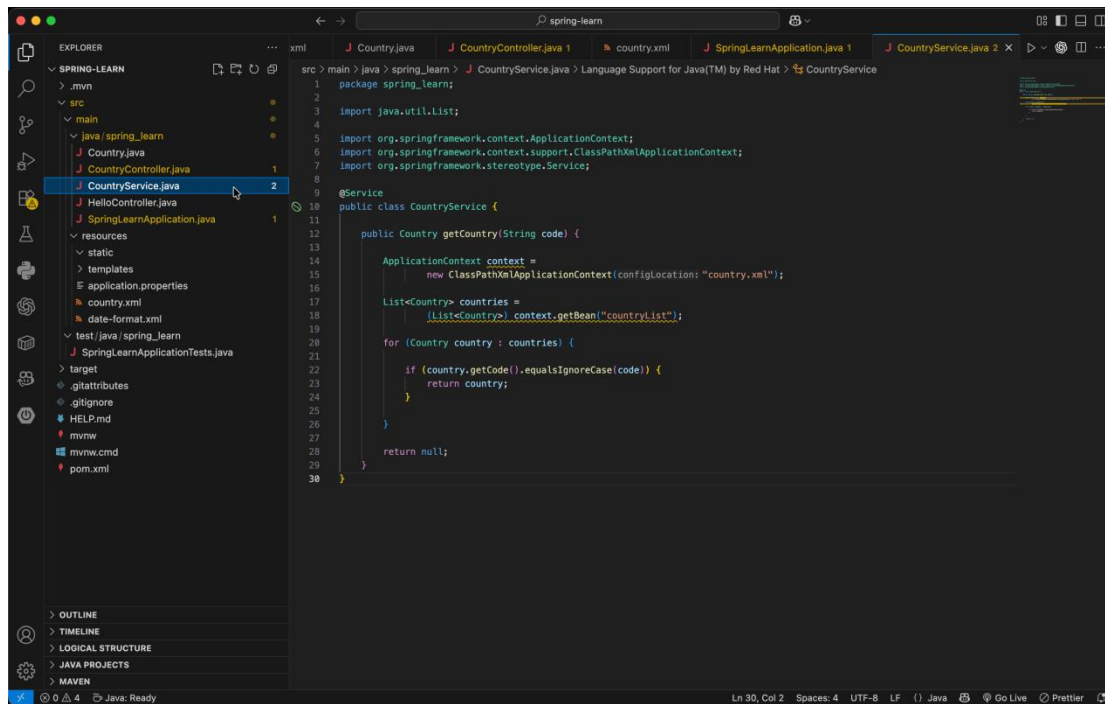
```
localhost:8083/country

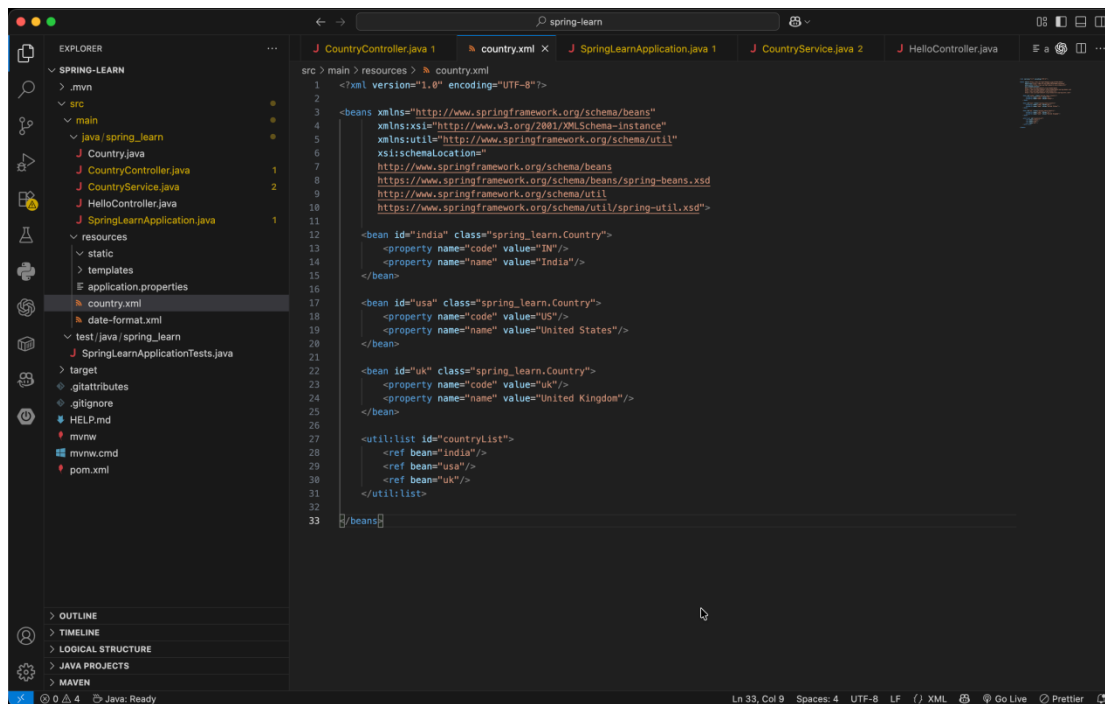
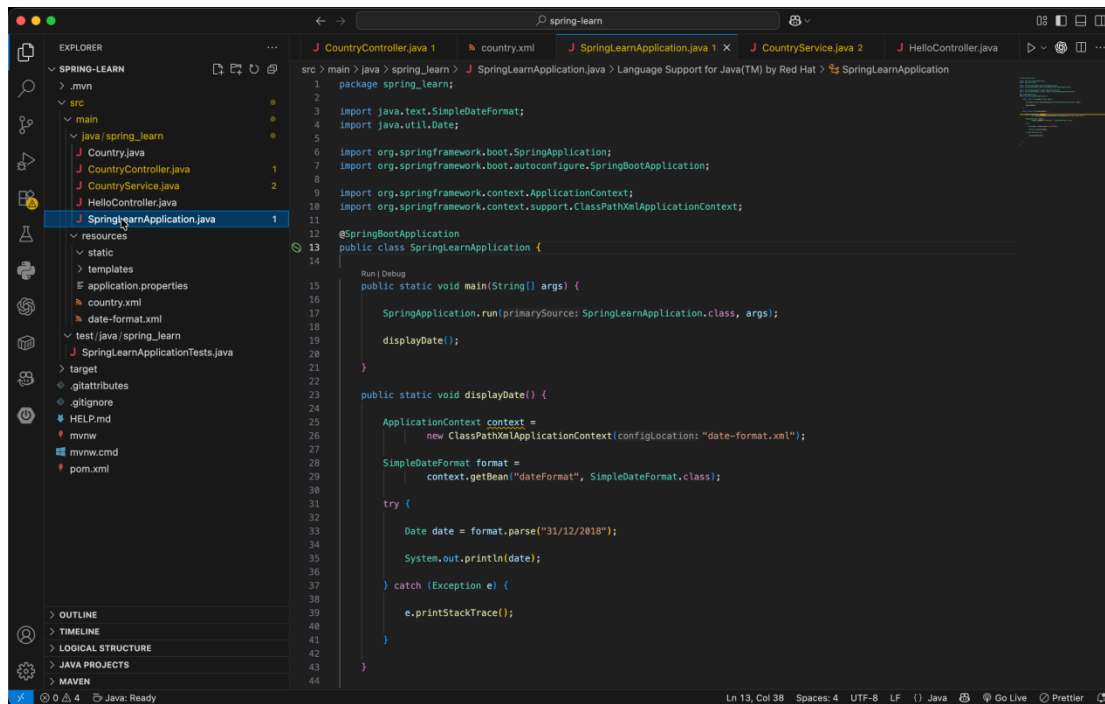
Pretty print

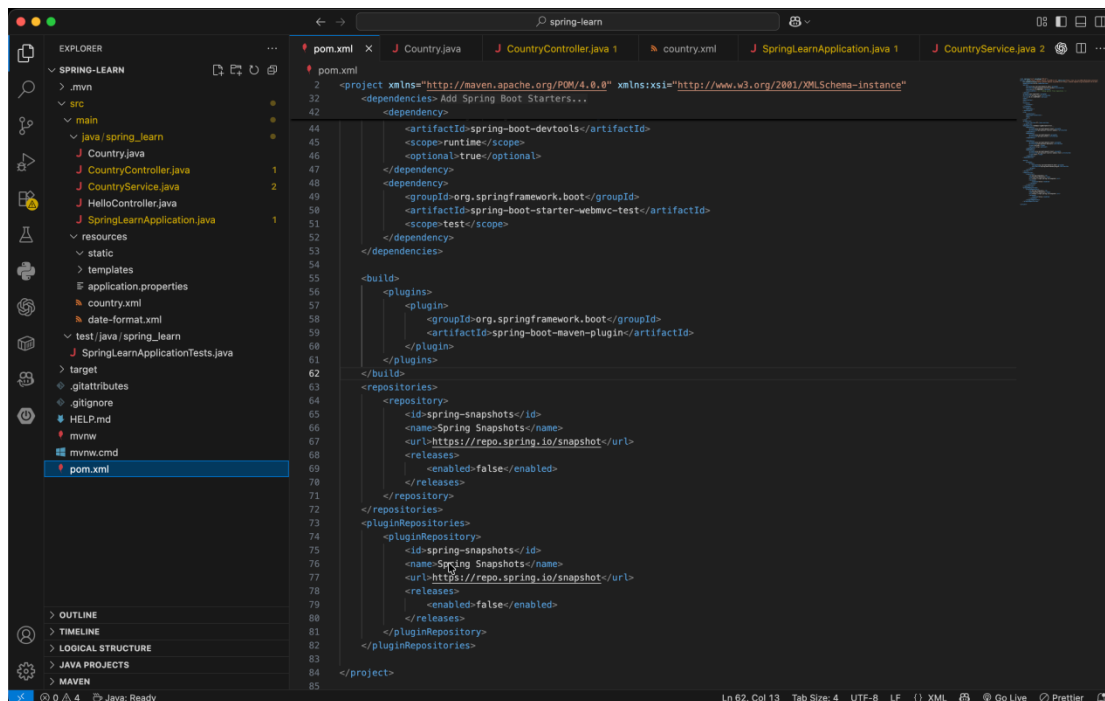
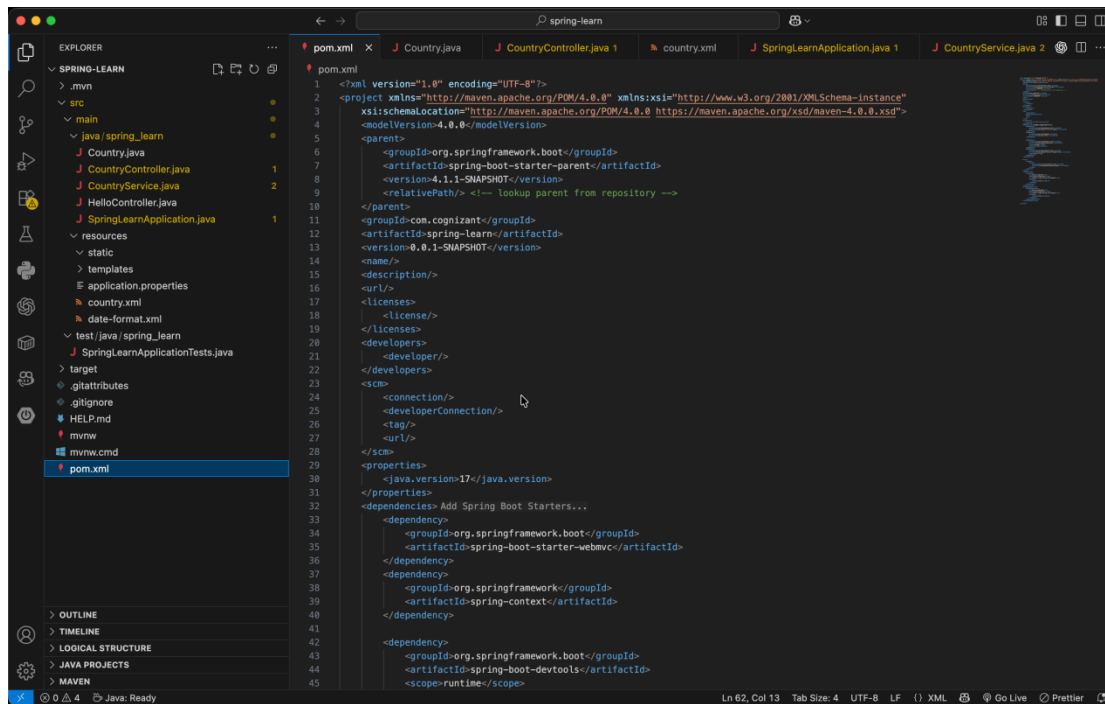
{"code": "IN", "name": "India"}
```

REST - Get country based on country code









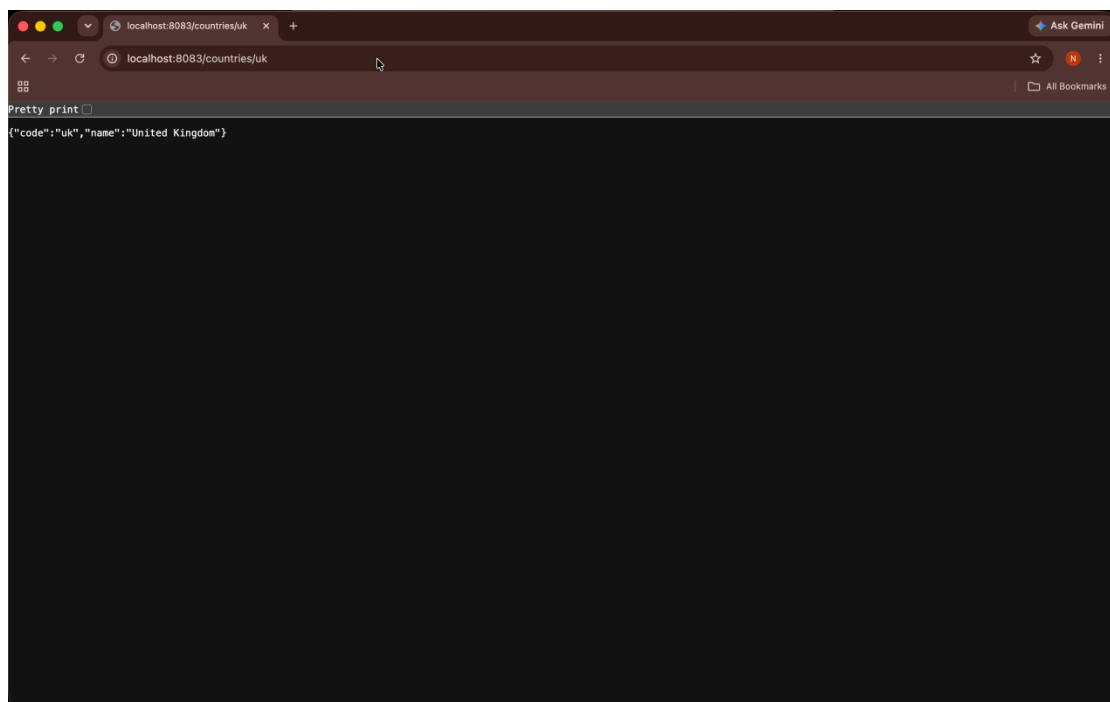
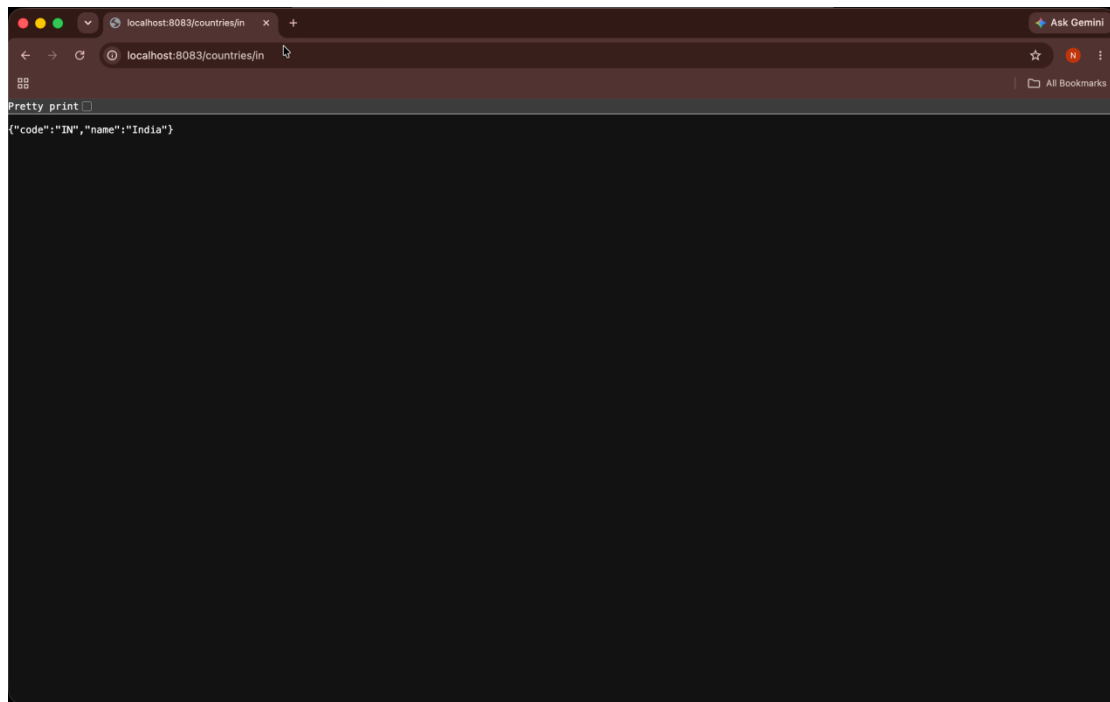
OUTPUT

```
[INFO] Copying 2 resources from src/main/resources to target/classes
[INFO] --- compiler:3.15.0:compile (default-compile) @ spring-learn ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 5 source files with javac [debug parameters release 17] to target/classes
[INFO] /Users/nithishvar/Downloads/spring-learn/src/main/java/spring_learn/CountryService.java: /Users/nithishvar/Downloads/spring-learn/src/main/java/spring_learn/CountryService.java uses unchecked or unsafe operations.
[INFO] /Users/nithishvar/Downloads/spring-learn/src/main/java/spring_learn/CountryService.java: Recompile with -Xlint:unchecked for details.
[INFO] --- resources:3.5.0:testResources (default-testResources) @ spring-learn ---
[INFO] skip non existing resourceDirectory /Users/nithishvar/Downloads/spring-learn/src/test/resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ spring-learn ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to target/test-classes
[INFO] <<< spring-boot:4.1.1-SNAPSHOT:run (default-cli) < test-compile @ spring-learn >>>
[INFO] --- spring-boot:4.1.1-SNAPSHOT:run (default-cli) @ spring-learn ---
[INFO] Attaching agents: []

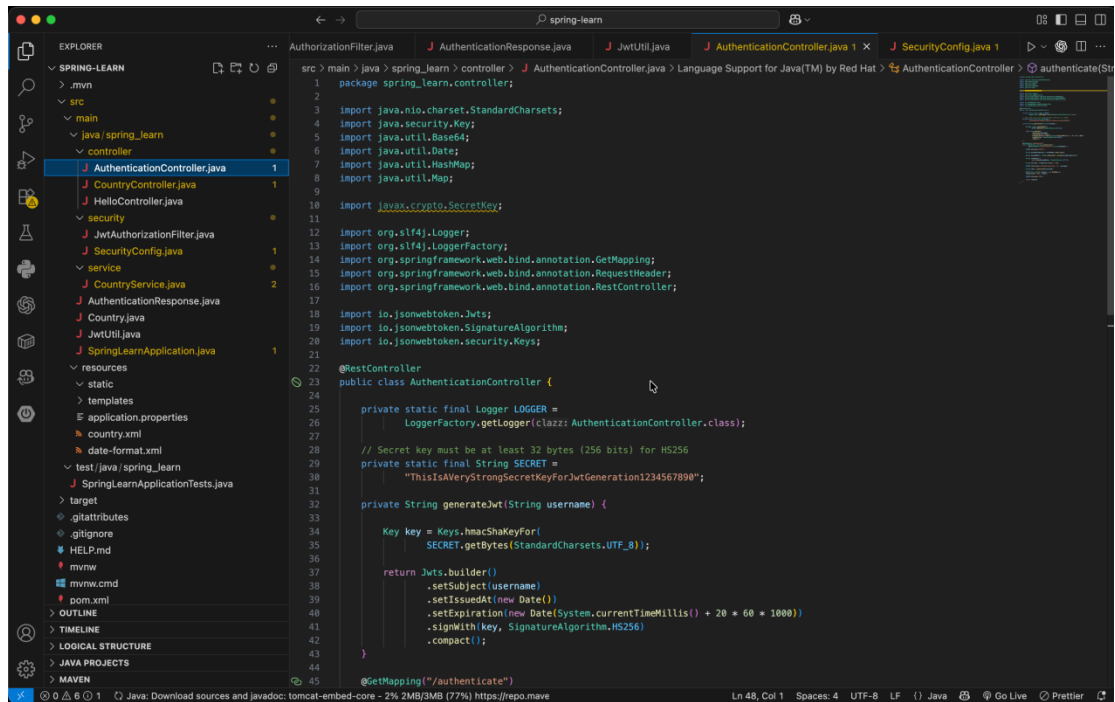
:: Spring Boot ::      (v4.1.1-SNAPSHOT)

2026-07-04T09:55:14.772+05:30 INFO 37564 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Starting SpringLearnApplication using Java 26.0.1 with PID 37564 (/Users/nithishvar/Downloads/spring-learn/target/classes started by nithishvar in /Users/nithishvar/Downloads/spring-learn)
2026-07-04T09:55:14.773+05:30 INFO 37564 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-04T09:55:14.862+05:30 INFO 37564 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults are active. Set 'spring.devtools.add-properties' to 'false' to disable
2026-07-04T09:55:14.882+05:30 INFO 37564 --- [spring-learn] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related details, see the 'spring.devtools.web' property in 'DEBUG'
2026-07-04T09:55:15.287+05:30 INFO 37564 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8083 (http)
2026-07-04T09:55:15.215+05:30 INFO 37564 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-04T09:55:15.216+05:30 INFO 37564 --- [spring-learn] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-04T09:55:15.232+05:30 INFO 37564 --- [spring-learn] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 429 ms
2026-07-04T09:55:15.448+05:30 INFO 37564 --- [spring-learn] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8083 (http) with context path '/'
2026-07-04T09:55:15.444+05:30 INFO 37564 --- [spring-learn] [ restartedMain] spring_learn.SpringLearnApplication : Started SpringLearnApplication in 0.867 seconds (process running for 1.084)
Mon Dec 31 00:00:00 IST 2018
2026-07-04T09:55:18.084+05:30 INFO 37564 --- [spring-learn] [nio-8083-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-04T09:55:18.084+05:30 INFO 37564 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-04T09:55:18.086+05:30 INFO 37564 --- [spring-learn] [nio-8083-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms
```

```
{\"code\": \"US\", \"name\": \"United States\"}
```

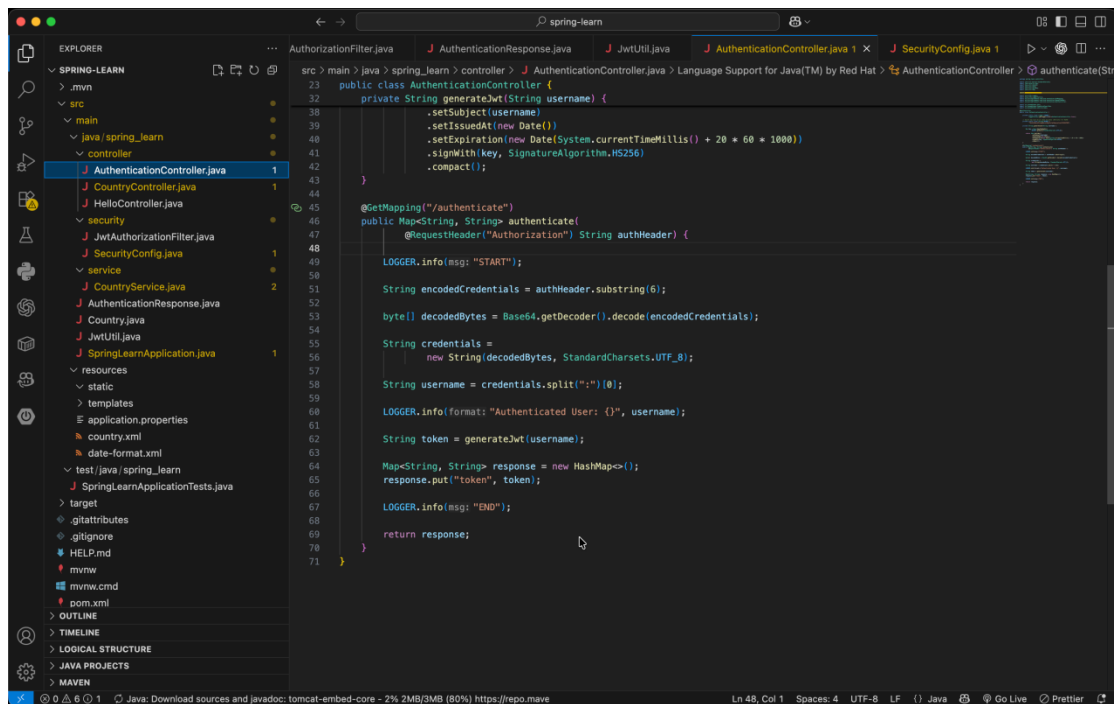


Create authentication service that returns JWT



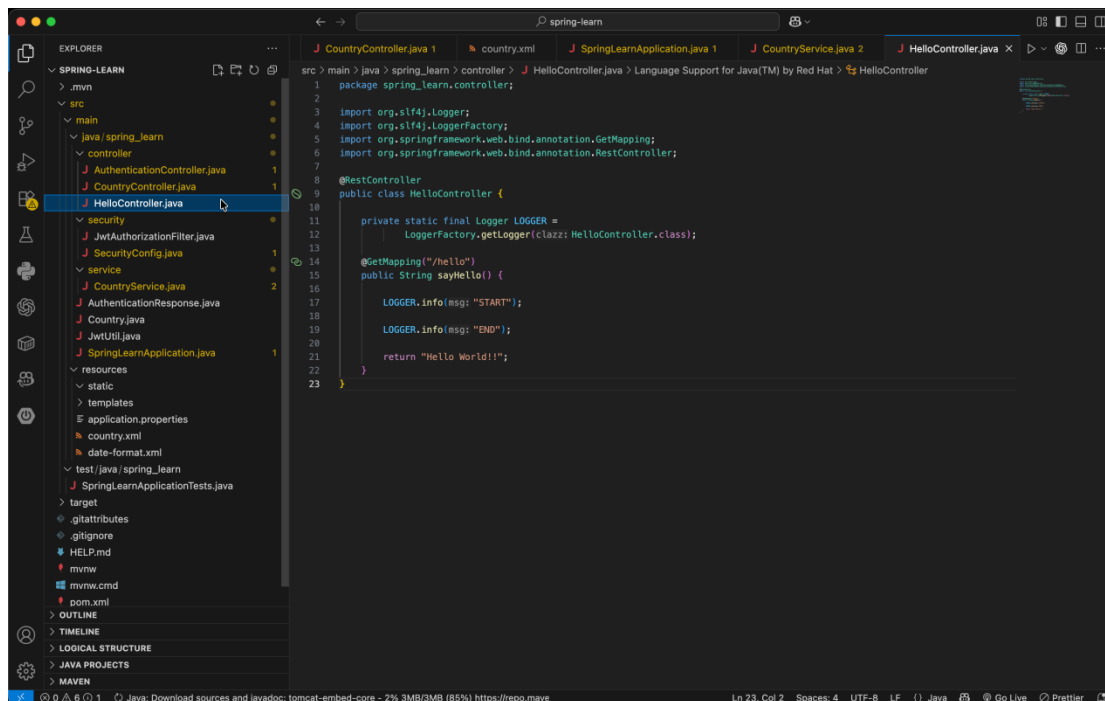
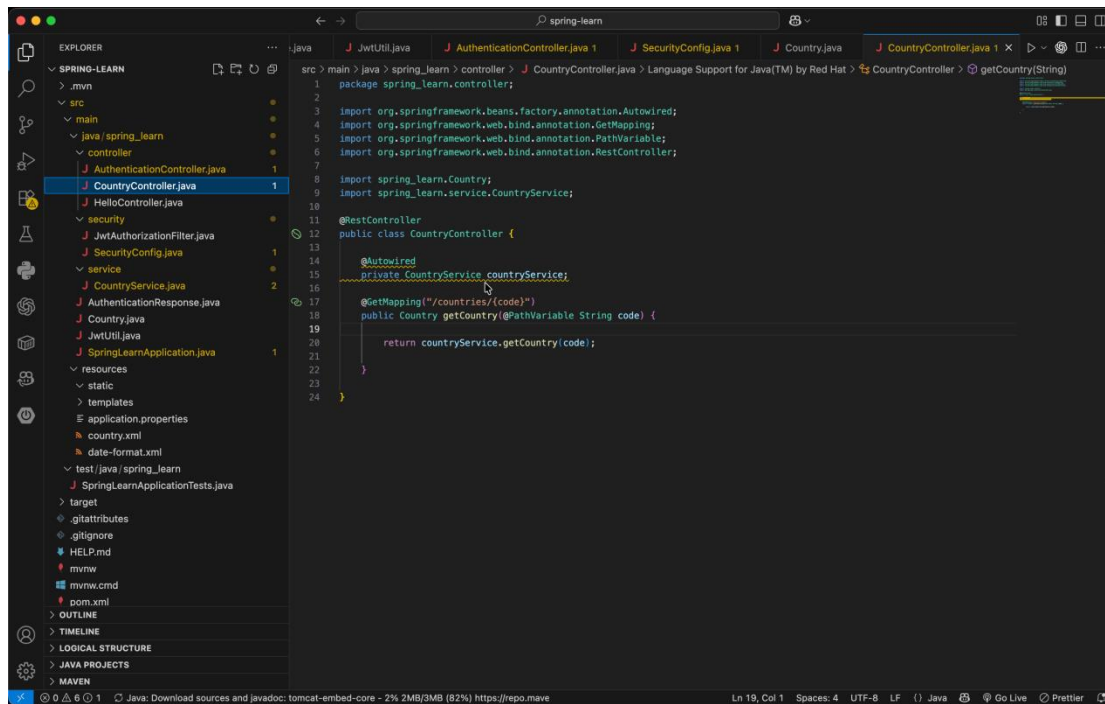
```
src > main > java > spring_learn > controller > J AuthenticationController.java > Language Support for Java(TM) by Red Hat > AuthenticationController > authenticate(Str

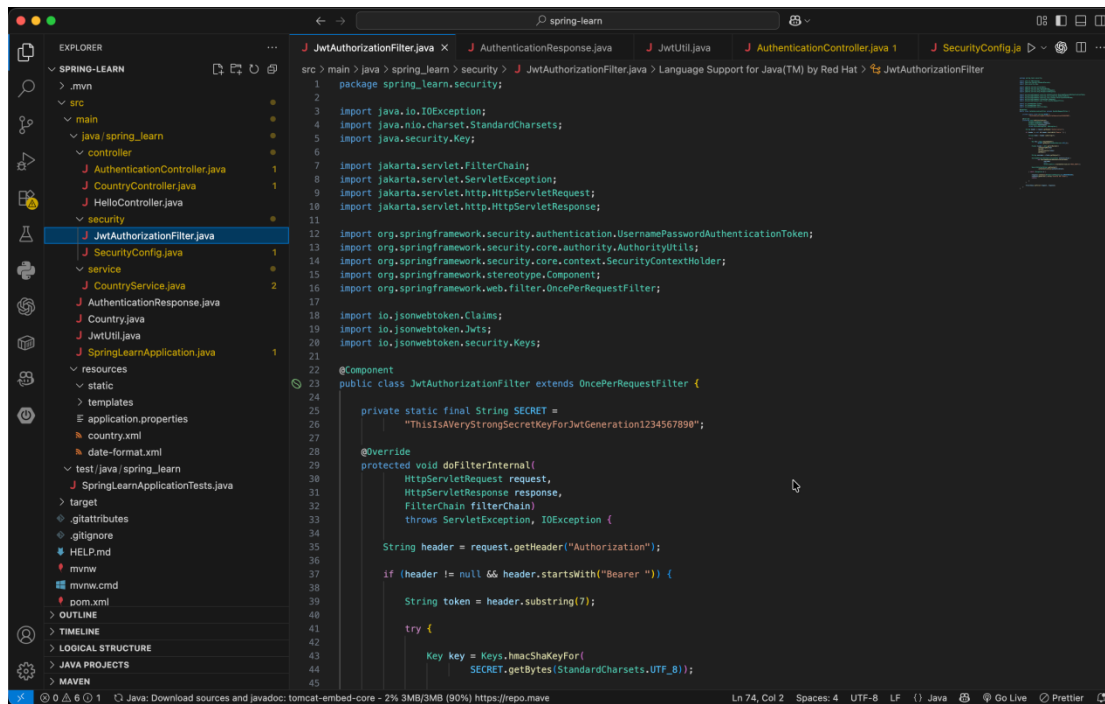
1 package spring_learn.controller;
2
3 import java.nio.charset.StandardCharsets;
4 import java.security.Key;
5 import java.util.Base64;
6 import java.util.Date;
7 import java.util.HashMap;
8 import java.util.Map;
9
10 import javax.crypto.SecretKey;
11
12 import org.slf4j.Logger;
13 import org.slf4j.LoggerFactory;
14 import org.springframework.web.bind.annotation.GetMapping;
15 import org.springframework.web.bind.annotation.RequestHeader;
16 import org.springframework.web.bind.annotation.RestController;
17
18 import io.jsonwebtoken.Jwts;
19 import io.jsonwebtoken.SignatureAlgorithm;
20 import io.jsonwebtoken.security.Keys;
21
22 @RestController
23 public class AuthenticationController {
24
25     private static final Logger LOGGER =
26         LoggerFactory.getLogger(clazz: AuthenticationController.class);
27
28     // Secret key must be at least 32 bytes (256 bits) for HS256
29     private static final String SECRET =
30         "ThisIsAVeryStrongSecretKeyForJwtGeneration1234567890";
31
32     private String generateJwt(String username) {
33
34         Key key = Keys.hmacShaKeyFor(
35             SECRET.getBytes(StandardCharsets.UTF_8));
36
37         return Jwts.builder()
38             .setSubject(username)
39             .setIssuedAt(new Date())
40             .setExpiration(new Date(System.currentTimeMillis() + 20 * 60 * 1000))
41             .signWith(key, SignatureAlgorithm.HS256)
42             .compact();
43     }
44
45     @GetMapping("/authenticate")
```



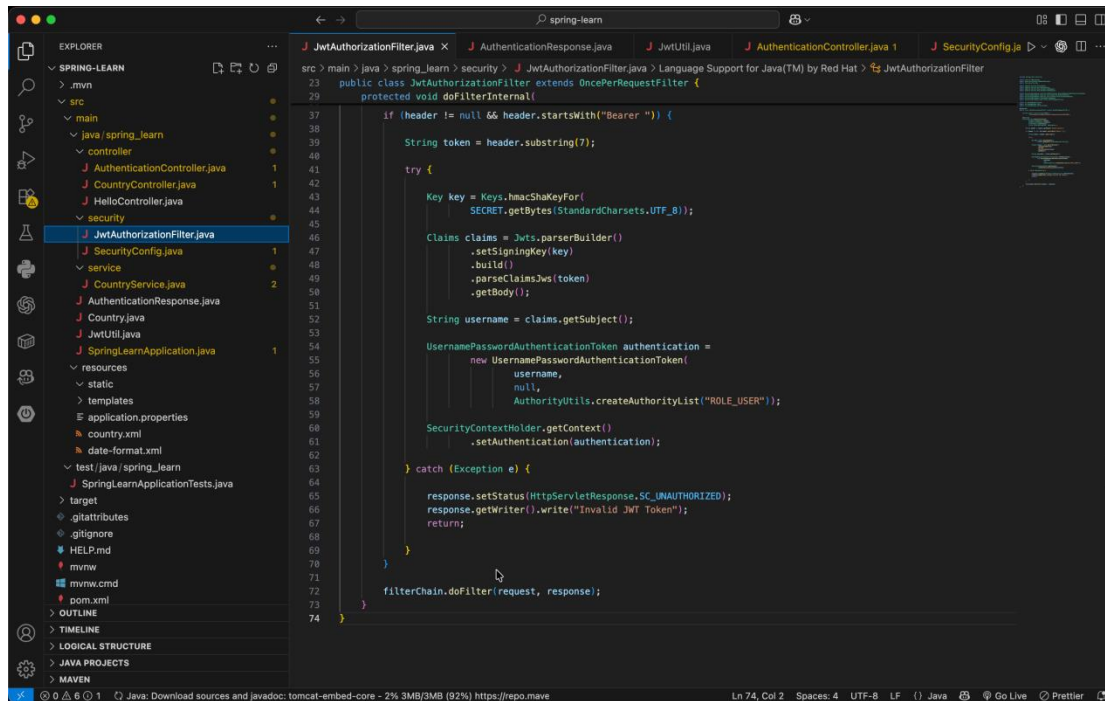
```
src > main > java > spring_learn > controller > J AuthenticationController.java > Language Support for Java(TM) by Red Hat > AuthenticationController > authenticate(Str

23 public class AuthenticationController {
24     private String generateJwt(String username) {
25         .setSubject(username)
26         .setIssuedAt(new Date())
27         .setExpiration(new Date(System.currentTimeMillis() + 20 * 60 * 1000))
28         .signWith(key, SignatureAlgorithm.HS256)
29         .compact();
30     }
31
32     @GetMapping("/authenticate")
33     public Map<String, String> authenticate(
34         @RequestHeader("Authorization") String authHeader) {
35
36         LOGGER.info(msg: "START");
37
38         String encodedCredentials = authHeader.substring(6);
39
40         byte[] decodedBytes = Base64.getDecoder().decode(encodedCredentials);
41
42         String credentials =
43             new String(decodedBytes, StandardCharsets.UTF_8);
44
45         String username = credentials.split(":")[0];
46
47         LOGGER.info(format: "Authenticated User: {}", username);
48
49         String token = generateJwt(username);
50
51         Map<String, String> response = new HashMap<>();
52         response.put("token", token);
53
54         LOGGER.info(msg: "END");
55
56         return response;
57     }
58 }
```





```
src > main > java > spring_learn > security > Jwtauthorizationfilter.java > Language Support for Java(TM) by Red Hat > Jwtauthorizationfilter
1 package spring_learn.security;
2
3 import java.io.IOException;
4 import java.nio.charset.StandardCharsets;
5 import java.security.Key;
6
7 import jakarta.servlet.FilterChain;
8 import jakarta.servlet.ServletException;
9 import jakarta.servlet.http.HttpServletRequest;
10 import jakarta.servlet.http.HttpServletResponse;
11
12 import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
13 import org.springframework.security.core.authority.AuthorityUtils;
14 import org.springframework.security.core.context.SecurityContextHolder;
15 import org.springframework.stereotype.Component;
16 import org.springframework.web.filter.OncePerRequestFilter;
17
18 import io.jsonwebtoken.Claims;
19 import io.jsonwebtoken.Jwts;
20 import io.jsonwebtoken.security.Keys;
21
22 @Component
23 public class JwtAuthorizationFilter extends OncePerRequestFilter {
24
25     private static final String SECRET =
26         "ThisIsAVeryStrongSecretKeyForJwtGeneration1234567890";
27
28     @Override
29     protected void doFilterInternal(
30         HttpServletRequest request,
31         HttpServletResponse response,
32         FilterChain filterChain)
33         throws ServletException, IOException {
34
35         String header = request.getHeader("Authorization");
36
37         if (header != null && header.startsWith("Bearer ")) {
38
39             String token = header.substring(7);
40
41             try {
42
43                 Key key = Keys.hmacShaKeyFor(
44                     SECRET.getBytes(StandardCharsets.UTF_8));
45
46                 Claims claims = Jwts.parserBuilder()
47                     .setSigningKey(key)
48                     .build()
49                     .parseClaimsJws(token)
50                     .getBody();
51
52                 String username = claims.getSubject();
53
54                 UsernamePasswordAuthenticationToken authentication =
55                     new UsernamePasswordAuthenticationToken(
56                         username,
57                         null,
58                         AuthorityUtils.createAuthorityList("ROLE_USER"));
59
60                 SecurityContextHolder.getContext().
61                     setAuthentication(authentication);
62
63             } catch (Exception e) {
64
65                 response.setStatus(HttpServletResponse.SC_UNAUTHORIZED);
66                 response.getWriter().write("Invalid JWT Token");
67                 return;
68             }
69
70             filterChain.doFilter(request, response);
71         }
72     }
73 }
74
```



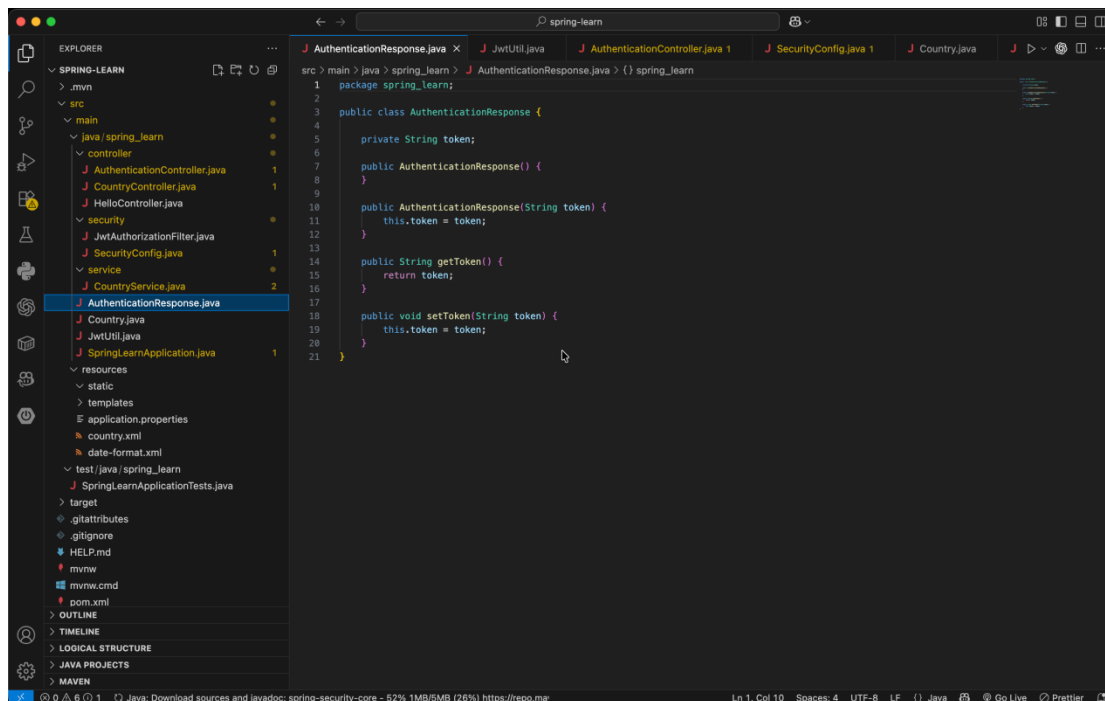
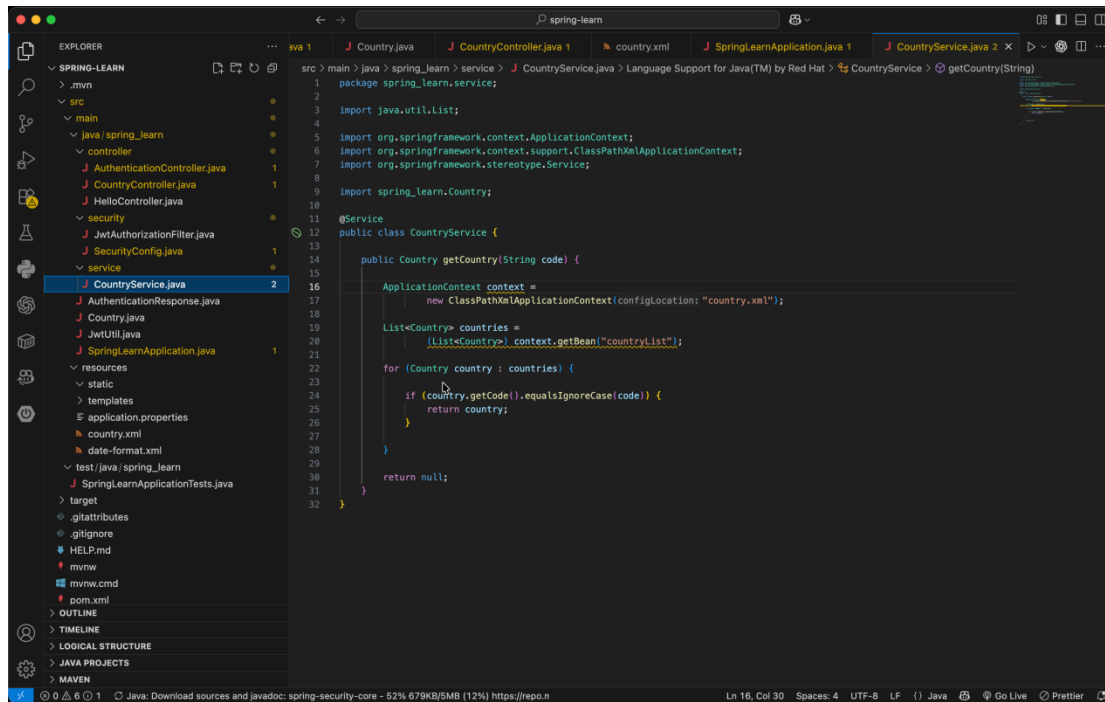
```
src > main > java > spring_learn > security > Jwtauthorizationfilter.java > Language Support for Java(TM) by Red Hat > Jwtauthorizationfilter
23 public class JwtAuthorizationFilter extends OncePerRequestFilter {
29     protected void doFilterInternal(
30
31         if (header != null && header.startsWith("Bearer ")) {
32
33             String token = header.substring(7);
34
35             try {
36
37                 Key key = Keys.hmacShaKeyFor(
38                     SECRET.getBytes(StandardCharsets.UTF_8));
39
40                 Claims claims = Jwts.parserBuilder()
41                     .setSigningKey(key)
42                     .build()
43                     .parseClaimsJws(token)
44                     .getBody();
45
46                 String username = claims.getSubject();
47
48                 UsernamePasswordAuthenticationToken authentication =
49                     new UsernamePasswordAuthenticationToken(
50                         username,
51                         null,
52                         AuthorityUtils.createAuthorityList("ROLE_USER"));
53
54                 SecurityContextHolder.getContext().
55                     setAuthentication(authentication);
56
57             } catch (Exception e) {
58
59                 response.setStatus(HttpServletResponse.SC_UNAUTHORIZED);
60                 response.getWriter().write("Invalid JWT Token");
61                 return;
62             }
63
64             filterChain.doFilter(request, response);
65         }
66     }
67 }
68
```

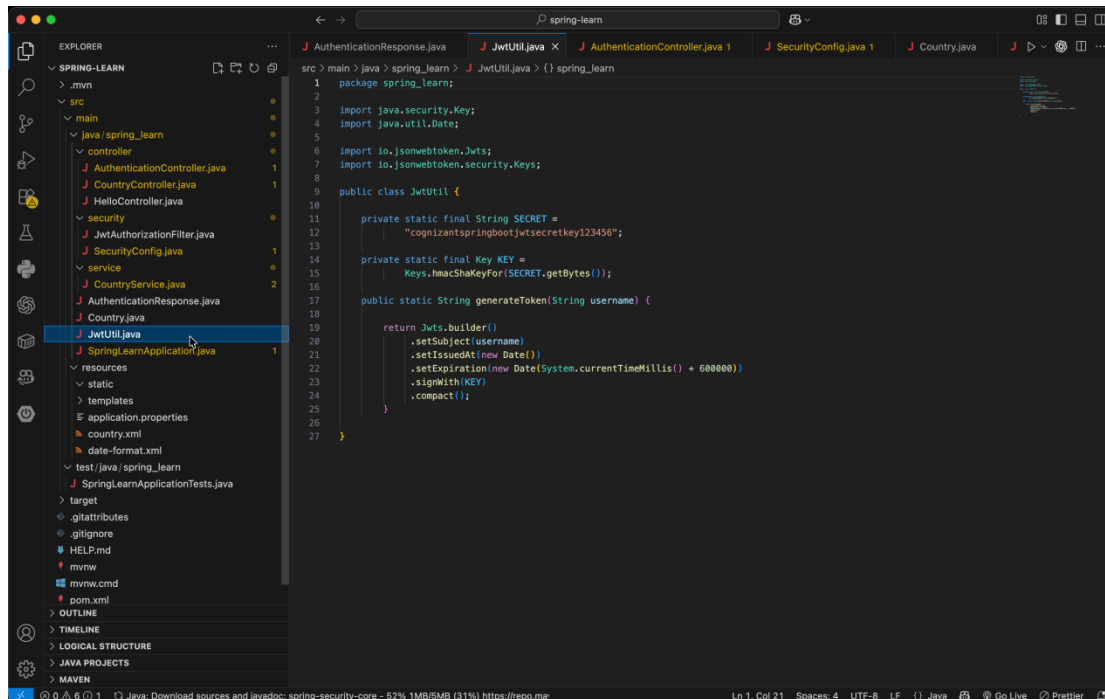
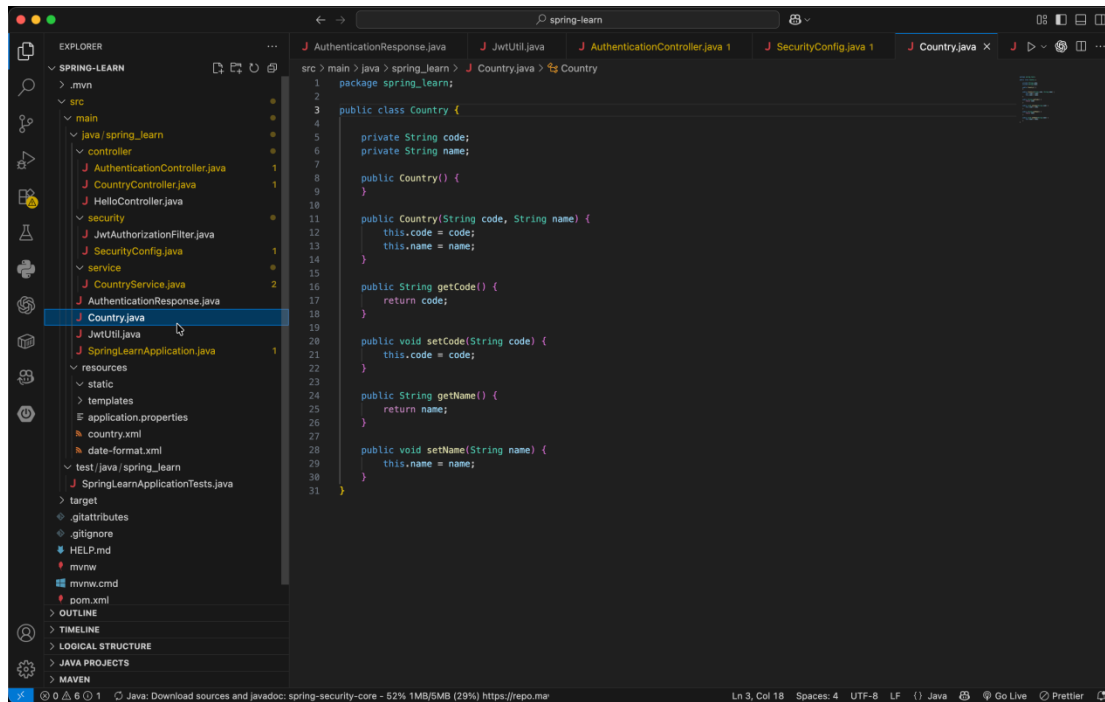
```
src > main > java > spring_learn > security > J SecurityConfig.java > Language Support for Java(TM) by Red Hat > SecurityConfig > filterChain(HttpSecurity)

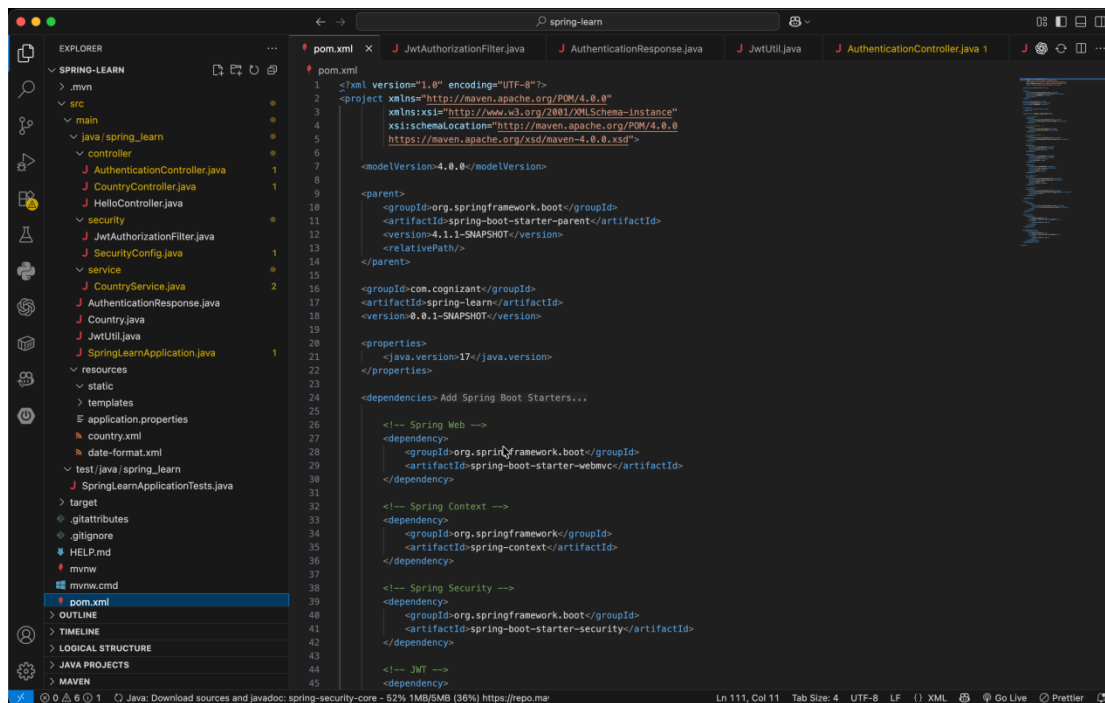
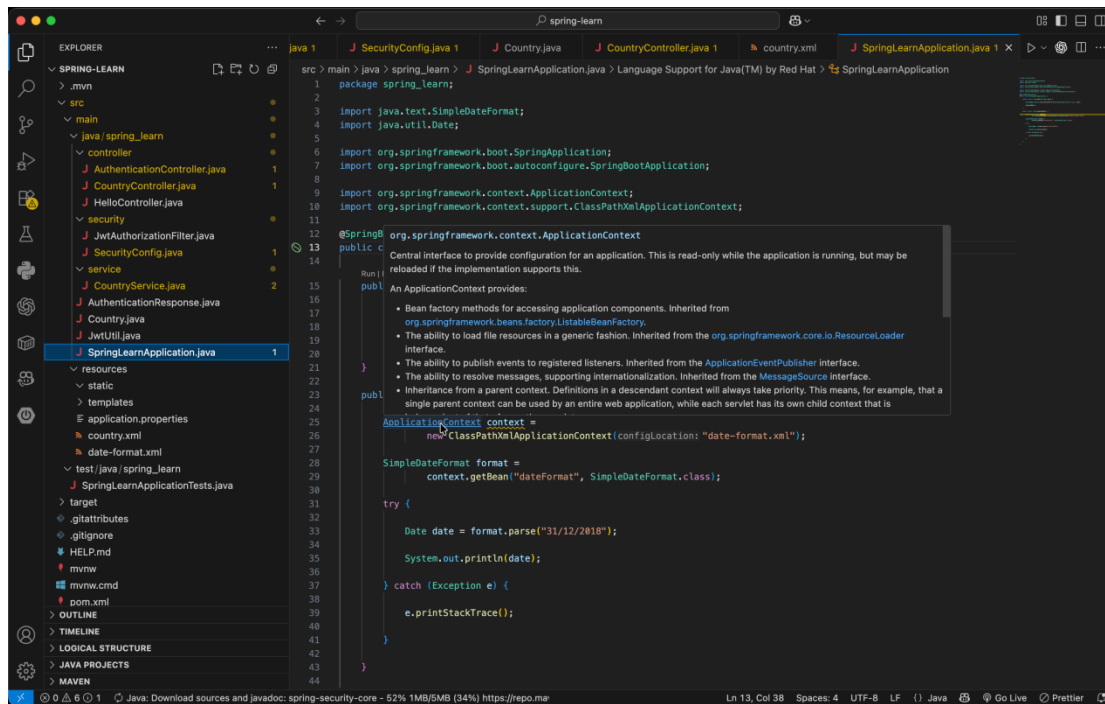
1 package spring_learn.security;
2 import org.springframework.beans.factory.annotation.Autowired;
3 import org.springframework.context.annotation.Bean;
4 import org.springframework.context.annotation.Configuration;
5 import org.springframework.security.config.annotation.web.builders.HttpSecurity;
6 import org.springframework.security.config.annotation.web.configuration.WebSecurityConfigurerAdapter;
7 import org.springframework.security.core.userdetails.User;
8 import org.springframework.security.core.userdetails.UserDetails;
9 import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
10 import org.springframework.security.crypto.password.PasswordEncoder;
11 import org.springframework.security.provisioning.InMemoryUserDetailsManager;
12 import org.springframework.security.web.SecurityFilterChain;
13 import org.springframework.security.web.authentication.UsernamePasswordAuthenticationFilter;
14 @Configuration
15 public class SecurityConfig {
16     @Autowired
17     private JwtauthorizationFilter jwtauthorizationFilter;
18     @Bean
19     public InMemoryUserDetailsManager userDetailsService() {
20         UserDetails admin = User.withUsername("admin")
21             .password(passwordEncoder().encode("pwd"))
22             .roles("ADMIN")
23             .build();
24         UserDetails user = User.withUsername("user")
25             .password(passwordEncoder().encode("pwd"))
26             .roles("USER")
27             .build();
28         return new InMemoryUserDetailsManager(admin, user);
29     }
30     @Bean
31     public PasswordEncoder passwordEncoder() {
32         return new BCryptPasswordEncoder();
33     }
34     @Bean
35     public SecurityFilterChain filterChain(HttpSecurity http) throws Exception {
36         http
37             .csrf(csrf -> csrf.disable())
38             .httpBasic(Customizer.withDefaults())
39             .authorizeHttpRequests(auth -> auth
40                 // Authentication API
41                 .requestMatchers(...patterns: "/authenticate").authenticated()
42                 // Protected APIs
43                 .requestMatchers(...patterns: "/countries/**").authenticated()
44                 // Public APIs
45                 .requestMatchers(...patterns: "/hello").permitAll()
46                 .requestMatchers(...patterns: "/country").permitAll()
47                 .anyRequest().permitAll()
48             )
49             .addFilterBefore(jwtauthorizationFilter,
50                 beforeFilter: UsernamePasswordAuthenticationFilter.class);
51         return http.build();
52     }
53 }
54
```

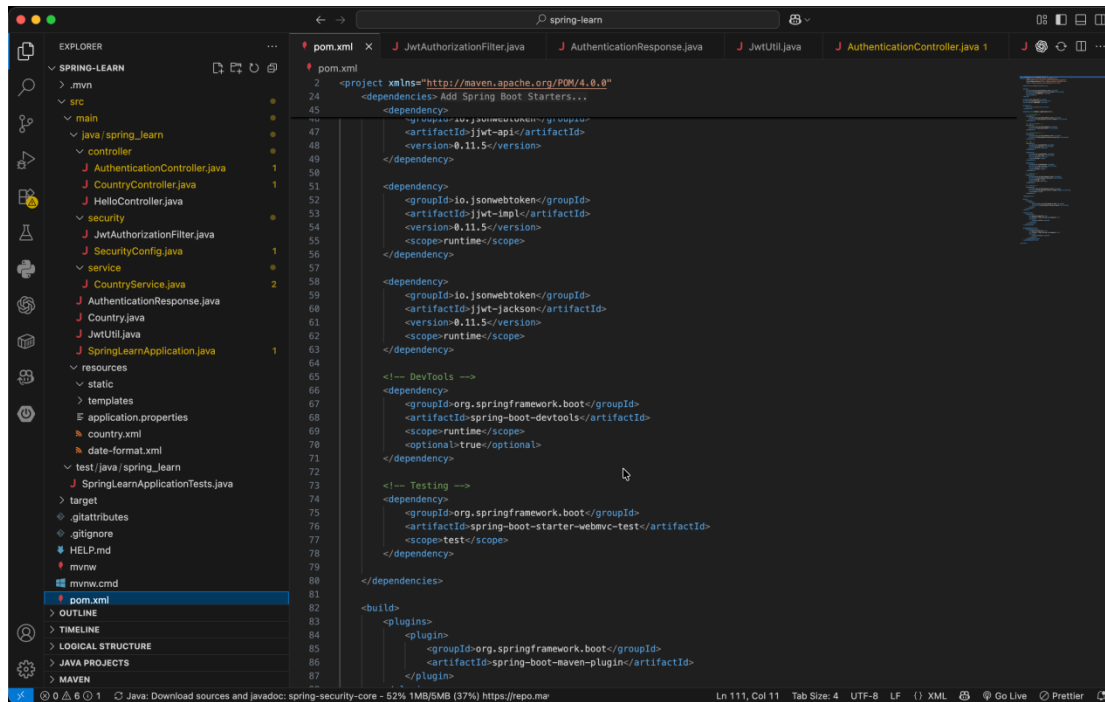
```
src > main > java > spring_learn > security > J SecurityConfig.java > Language Support for Java(TM) by Red Hat > SecurityConfig > filterChain(HttpSecurity)

15 public class SecurityConfig {
16     @Bean
17     public InMemoryUserDetailsManager userDetailsService() {
18         UserDetails admin = User.withUsername("admin")
19             .password(passwordEncoder().encode("pwd"))
20             .roles("ADMIN")
21             .build();
22         UserDetails user = User.withUsername("user")
23             .password(passwordEncoder().encode("pwd"))
24             .roles("USER")
25             .build();
26         return new InMemoryUserDetailsManager(admin, user);
27     }
28     @Bean
29     public PasswordEncoder passwordEncoder() {
30         return new BCryptPasswordEncoder();
31     }
32     @Bean
33     public SecurityFilterChain filterChain(HttpSecurity http) throws Exception {
34         http
35             .csrf(csrf -> csrf.disable())
36             .httpBasic(Customizer.withDefaults())
37             .authorizeHttpRequests(auth -> auth
38                 // Authentication API
39                 .requestMatchers(...patterns: "/authenticate").authenticated()
40                 // Protected APIs
41                 .requestMatchers(...patterns: "/countries/**").authenticated()
42                 // Public APIs
43                 .requestMatchers(...patterns: "/hello").permitAll()
44                 .requestMatchers(...patterns: "/country").permitAll()
45                 .anyRequest().permitAll()
46             )
47             .addFilterBefore(jwtauthorizationFilter,
48                 beforeFilter: UsernamePasswordAuthenticationFilter.class);
49         return http.build();
50     }
51 }
52
```









[illegible]

The screenshot displays an IDE interface for a Spring Boot project named 'spring-learn'. The Explorer view on the left shows the project structure, including files like AuthenticationController.java, CountryController.java, HelloController.java, SecurityConfig.java, and SpringLearnApplicationTests.java. The Terminal view on the right shows the command prompt output, including the error message: 'Error while loading conda entry point: anaconda-cloud-auth (cannot import name 'AliasGenerator' from 'pydantic')'. The status bar at the bottom indicates 'Ln 65, Col 35 Spaces: 4 UTF-8 LF'.

